

ARM PrimeCell™ Synchronous Serial Port (PL022) Technical Reference Manual

ARM

ARM PrimeCell Synchronous Serial Port (PL022)

Technical Reference Manual

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Release Information

Change history

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Preface

This preface introduces the ARM PrimeCell Synchronous Serial Port (PL022) and its reference documentation. It contains the following sections:

- *About this document* on page vi
- *Further reading* on page viii
- *Feedback* on page ix.

About this document

This document is the ARM PrimeCell Synchronous Serial Port (PL022) Technical Reference Manual.

Intended audience

This document has been written for hardware and software engineers implementing *System-on-Chip* (SoC) designs. It provides the necessary information to enable designers to integrate the peripheral into a target system as quickly as possible.

Using this manual

This document is organized into the following chapters:

Chapter 1 *Introduction*

Read this chapter for an introduction to the PrimeCell *Synchronous Serial Port* (SSP) and its features.

Chapter 2 *Functional Overview*

Read this chapter for a description of the major functional blocks of the PrimeCell SSP.

Chapter 3 *Programmer's Model*

Read this chapter for a description of the PrimeCell SSP registers and programming details.

Chapter 4 *Programmer's Model for Test*

Read this chapter for a description of the logic in the PrimeCell SSP for integration testing.

Appendix A *Signal Descriptions*

Read this appendix for details of the PrimeCell SSP signals.

Typographical conventions

The following typographical conventions are used in this book:

<i>italic</i>	Highlights important notes, introduces special terminology, denotes internal cross-references, and citations.
bold	Highlights interface elements, such as menu names. Denotes ARM processor signal names. Also used for terms in descriptive lists, where appropriate.

<code>monospace</code>	Denotes text that can be entered at the keyboard, such as commands, file and program names, and source code.
<u><code>monospace</code></u>	Denotes a permitted abbreviation for a command or option. The underlined text can be entered instead of the full command or option name.
<i><code>monospace italic</code></i>	Denotes arguments to commands and functions where the argument is to be replaced by a specific value.
<code>monospace bold</code>	Denotes language keywords when used outside example code.

Timing diagram conventions

This manual contains one or more timing diagrams. Figure Pre-1 explains the components used in these diagrams. Any variations are clearly labeled when they occur. Therefore, no additional meaning should be attached unless specifically stated.

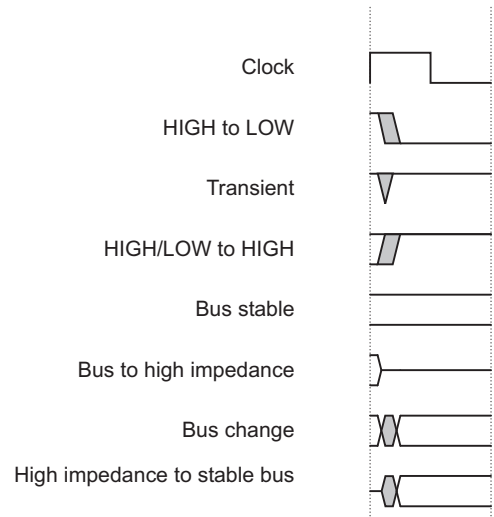


Figure Pre-1 Key to timing diagram conventions

Shaded bus and signal areas are undefined, so the bus or signal can assume any value within the shaded area at that time. The actual level is unimportant and does not affect normal operation.

Further reading

This section lists publications by ARM Limited, and by third parties.

ARM periodically provides updates and corrections to its documentation. See <http://www.arm.com> for current errata sheets and addenda.

See also the ARM Frequently Asked Questions list at:
<http://www.arm.com/DevSupp/Sales+Support/faq.html>

ARM publications

This document contains information that is specific to the PrimeCell Synchronous Serial Port. Refer to the following documents for other relevant information:

- *AMBA Specification (Rev 2.0)* (ARM IHI 00011)
- *ARM PrimeCell Synchronous Serial Port (PL022) Integration Manual* (PL022 INTM 0000)
- *ARM PrimeCell Synchronous Serial Port (PL022) Design Manual* (PL022 DDES 0000)
- *ARM PrimeCell Synchronous Serial Port (PL020) Technical Reference Manual* (ARM DDI 0141)
- *ARM PrimeCell Synchronous Serial Port Master and Slave (PL021) Technical Reference Manual* (ARM DDI 0171).

Feedback

ARM welcomes feedback both on the ARM PrimeCell Synchronous Serial Port (PL022), and on the documentation.

Feedback on the ARM PrimeCell SSP (PL022)

If you have any comments or suggestions about this product, please contact your supplier giving:

- the product name
- a concise explanation of your comments.

Feedback on this document

If you have any comments on about this document, please send email to errata@arm.com giving:

- the document title
- the document number
- the page number(s) to which your comments refer
- a concise explanation of your comments.

General suggestions for additions and improvements are also welcome.

Chapter 1

Introduction

This chapter introduces the ARM PrimeCell Synchronous Serial Port (PL022). It contains the following section:

- *About the ARM PrimeCell SSP (PL022)* on page 1-2.

1.1 About the ARM PrimeCell SSP (PL022)

The PrimeCell *Synchronous Serial Port* (SSP) is an *Advanced Microcontroller Bus Architecture* (AMBA) slave block that connects to the *Advanced Peripheral Bus* (APB). The PrimeCell SSP is an AMBA compliant *System-on-Chip* (SoC) peripheral that is developed, tested, and licensed by ARM.

The PrimeCell SSP is a master or slave interface that enables synchronous serial communication with slave or master peripherals having one of the following:

- a Motorola SPI-compatible interface
- a Texas Instruments synchronous serial interface
- a National Semiconductor Microwire interface.

In both master and slave configurations, the PrimeCell SSP performs:

- parallel-to-serial conversion on data written to an internal 16-bit wide, 8-location deep transmit FIFO
- serial-to-parallel conversion on received data, buffering it in a similar 16-bit wide, 8-location deep receive FIFO.

Interrupts are generated to:

- request servicing of the transmit and receive FIFO
- inform the system that a receive FIFO over-run has occurred
- inform the system that data is present in the receive FIFO after an idle period has expired.

The features of the PrimeCell SSP are covered under the following headings:

- *Features of the PrimeCell SSP* on page 1-3
- *SPI features* on page 1-5
- *Microwire features* on page 1-5
- *Texas Instruments synchronous serial interface features* on page 1-5.

Note

Due to changes in the programmer's model, the PrimeCell SSP (PL022) is not backwards compatible with the previous PrimeCell SSPMS (PL021) or PrimeCell SSP (PL020).

1.1.1 Features of the PrimeCell SSP

The PrimeCell SSP has the following features:

- Compliance to the *AMBA Specification (Rev 2.0)* for easy integration into SoC implementation.
- Master or slave operation.
- Programmable clock bit rate and prescale.
- Separate transmit and receive first-in, first-out memory buffers, 16 bits wide, 8 locations deep.
- Programmable choice of interface operation, SPI, Microwire, or TI synchronous serial.
- Programmable data frame size from 4 to 16 bits.
- Independent masking of transmit FIFO, receive FIFO, and receive overrun interrupts.
- Internal loopback test mode available.
- Support for *Direct Memory Access* (DMA).
- Identification registers that uniquely identify the PrimeCell SSP. These can be used by an operating system to automatically configure itself.

Figure 1-1 on page 1-4 shows a block diagram of the PrimeCell SSP.

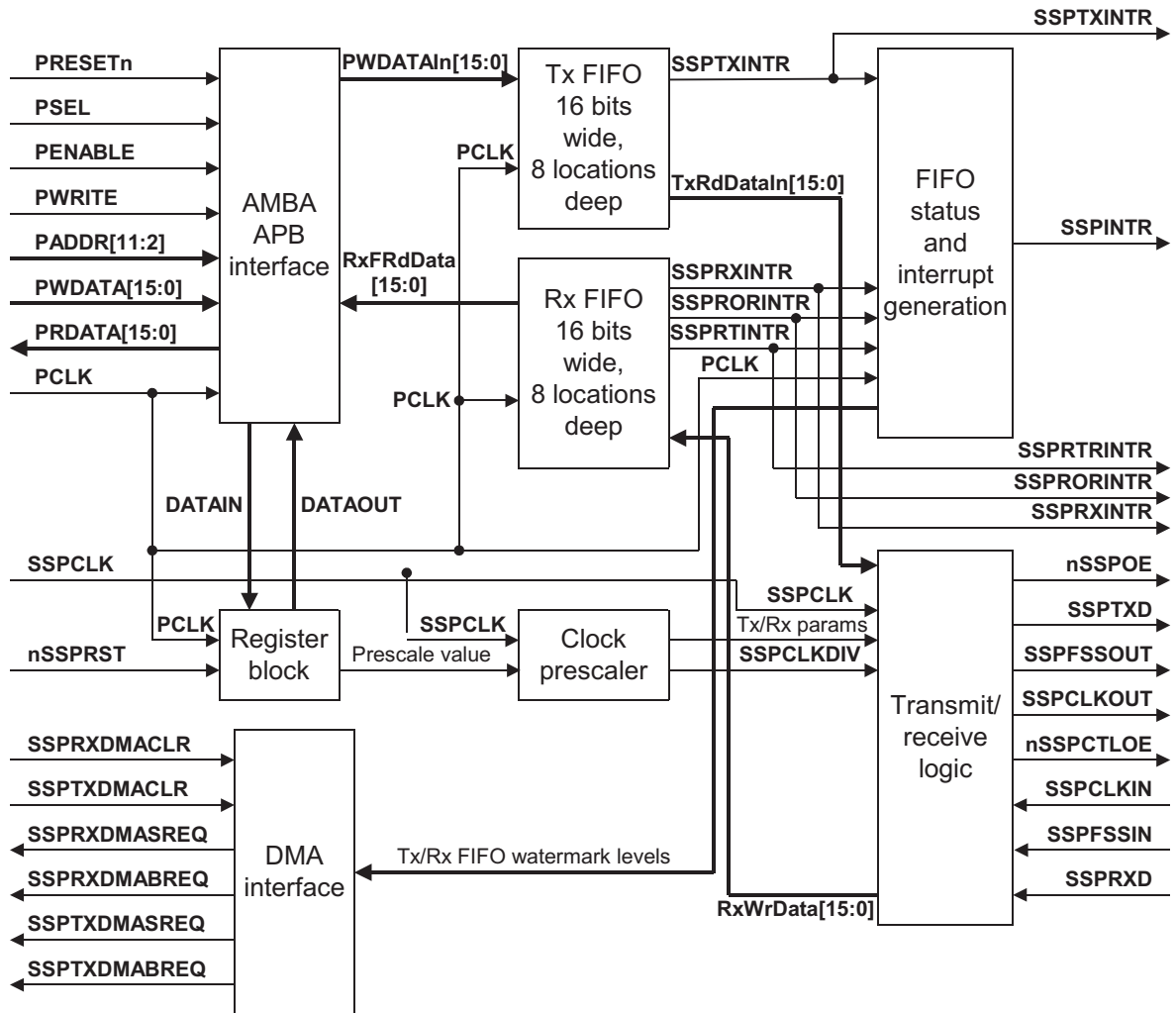


Figure 1-1 PrimeCell SSP block diagram

———— **Note** ————

For clarity, test logic is not represented in Figure 1-1.

1.1.2 Programmable parameters

The following parameters are programmable:

- master or slave mode
- enabling of operation
- frame format
- communication baud rate
- clock phase and polarity
- data widths from 4 to 16 bits wide
- interrupt masking.

1.1.3 SPI features

The features of the Motorola SPI-compatible interface are:

- full duplex, four-wire synchronous transfers
- programmable clock polarity and phase.

1.1.4 Microwire features

The features of the National Semiconductor Microwire interface are:

- half-duplex transfer using 8-bit control message.

1.1.5 Texas Instruments synchronous serial interface features

The features of the Texas Instruments synchronous serial interface are:

- full-duplex, four-wire synchronous transfer
- transmit data pin tristateable when not transmitting.

Chapter 2

Functional Overview

This chapter describes the major functional blocks of the ARM PrimeCell Synchronous Serial Port (PL022) and contains the following sections:

- *ARM PrimeCell SSP (PL022) overview* on page 2-2
- *PrimeCell SSP functional description* on page 2-3
- *PrimeCell SSP operation* on page 2-7.

2.1 ARM PrimeCell SSP (PL022) overview

The PrimeCell SSP is a master or slave interface for synchronous serial communication with peripheral devices that have either Motorola SPI, National Semiconductor Microwire or Texas Instruments synchronous serial interfaces.

The PrimeCell SSP performs serial-to-parallel conversion on data received from a peripheral device. The CPU accesses data, control, and status information through the AMBA APB interface. The transmit and receive paths are buffered with internal FIFO memories allowing up to eight 16-bit values to be stored independently in both transmit and receive modes. Serial data is transmitted on **SSPTXD** and received on **SSPRXD**.

The PrimeCell SSP includes a programmable bit rate clock divider and prescaler to generate the serial output clock **SSPCLKOUT** from the input clock **SSPCLK**. Bit rates are supported to 2MHz and higher, subject to choice of frequency for **SSPCLK** and the maximum bit rate is determined by peripheral devices.

The PrimeCell SSP operating mode, frame format, and size are programmed through the control registers **SSPCR0** and **SSPCR1**.

Four individually maskable interrupt outputs are generated:

- **SSPTXINTR** requests servicing of the transmit buffer
- **SSPRXINTR** requests servicing of the receive buffer
- **SSPRORINTR** indicates an overrun condition in the receive FIFO
- **SSPRTINTR** indicates that a timeout period expired while data was present in the receive FIFO.

A single combined interrupt, **SSPINTR** output, is asserted if any of the individual interrupts are asserted and unmasked.

In addition to the above interrupts, a set of DMA signals are provided for interfacing with a DMA controller.

Depending on the operating mode selected, the **SSPFSSOUT** output operates as an active HIGH frame synchronization output for Texas Instruments synchronous serial frame format or an active LOW slave select for SPI and Microwire.

2.2 PrimeCell SSP functional description

The PrimeCell SSP block diagram is shown in Figure 2-1.

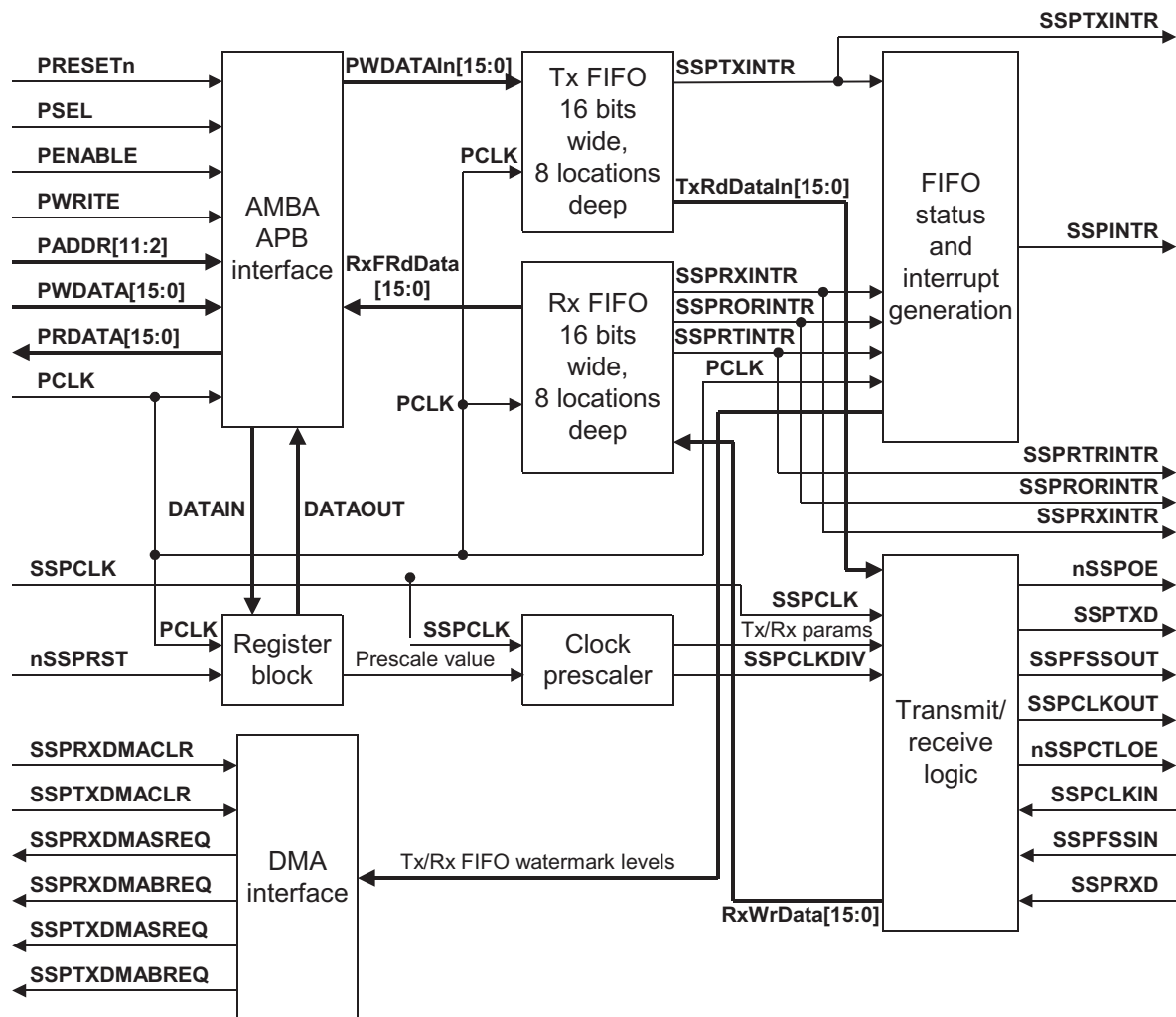


Figure 2-1 PrimeCell SSP block diagram

Note

For clarity, test logic is not represented in Figure 2-1.

The functions of the PrimeCell SSP are described in the following sections:

- *AMBA APB interface*
- *Register block*
- *Clock prescaler*
- *Transmit FIFO* on page 2-5
- *Receive FIFO* on page 2-5
- *Transmit and receive logic* on page 2-5
- *Interrupt generation logic* on page 2-5
- *Synchronizing registers and logic* on page 2-6
- *DMA interface* on page 2-6.

2.2.1 AMBA APB interface

The AMBA APB interface generates read and write decodes for accesses to status and control registers, and transmit and receive FIFO memories.

The AMBA APB is a local secondary bus that provides a low-power extension to the higher bandwidth AMBA *Advanced High-performance Bus* (AHB) within the AMBA system hierarchy. The AMBA APB groups narrow-bus peripherals to avoid loading the system bus and provides an interface using memory-mapped registers, which are accessed under programmed control.

2.2.2 Register block

The register block stores data written or to be read across the AMBA APB interface.

2.2.3 Clock prescaler

When configured as a master, an internal prescaler, comprising two free-running reloadable serially linked counters, is used to provide the serial output clock **SSPCLKOUT**.

You can program the clock prescaler, through the SSPCPSR register, to divide **SSPCLK** by a factor of 2 to 254 in steps of two. By not utilizing the least significant bit of the SSPCPSR register, division by an odd number is not possible and this ensures a symmetrical (equal mark space ratio) clock is generated.

The output of the prescaler is further divided by a factor of 1 to 256, through the programming of the SSPCR0 control register, to give the final master output clock **SSPCLKOUT**.

2.2.4 Transmit FIFO

The common transmit FIFO is a 16-bit wide, 8-locations deep, first-in, first-out memory buffer. CPU data written across the AMBA APB interface are stored in the buffer until read out by the transmit logic.

When configured as a master or a slave parallel data is written into the transmit FIFO prior to serial conversion and transmission to the attached slave or master respectively, through the **SSPTXD** pin.

2.2.5 Receive FIFO

The common receive FIFO is a 16-bit wide, 8-locations deep, first-in, first-out memory buffer. Received data from the serial interface are stored in the buffer until read out by the CPU across the AMBA APB interface.

When configured as a master or slave, serial data received through the **SSPRXD** pin is registered prior to parallel loading into the attached slave or master receive FIFO respectively.

2.2.6 Transmit and receive logic

When configured as a master, the clock to the attached slaves is derived from a divided down version of **SSPCLK** through the prescaler operations described previously. The master transmit logic successively reads a value from its transmit FIFO and performs parallel to serial conversion on it. Then the serial data stream and frame control signal, synchronized to **SSPCLKOUT**, are output through the **SSPTXD** pin to the attached slaves. The master receive logic performs serial to parallel conversion on the incoming synchronous **SSPRXD** data stream, extracting and storing values into its receive FIFO, for subsequent reading through the APB interface.

When configured as a slave, the **SSPCLKIN** clock is provided by an attached master and used to time its transmission and reception sequences. The slave transmit logic, under control of the master clock, successively reads a value from its transmit FIFO, performs parallel to serial conversion, then output the serial data stream and frame control signal through the slave **SSPTXD** pin. The slave receive logic performs serial to parallel conversion on the incoming **SSPRXD** data stream, extracting and storing values into its receive FIFO, for subsequent reading through the APB interface.

2.2.7 Interrupt generation logic

Four individual maskable, active HIGH interrupts are generated by the PrimeCell SSP. A combined interrupt output is also generated as an OR function of the individual interrupt requests.

You can use the single combined interrupt with a system interrupt controller that provides another level of masking on a per-peripheral basis. This allows use of modular device drivers that always know where to find the interrupt source control register bits.

The individual interrupt requests could also be used with a system interrupt controller that provides masking for the outputs of each peripheral. In this way, a global interrupt controller service routine would be able to read the entire set of sources from one wide register in the system interrupt controller. This is attractive where the time to read from the peripheral registers is significant compared to the CPU clock speed in a real-time system.

The peripheral supports both the above methods.

The transmit and receive dynamic data-flow interrupts, **SSPTXINTR** and **SSPRXINTR**, are separated from the status interrupts so that data can be read or written in response to the FIFO trigger levels.

2.2.8 DMA interface

The PrimeCell SSP provides an interface to connect to a DMA controller. See *PrimeCell DMA interface* on page 2-25 for details.

2.2.9 Synchronizing registers and logic

The PrimeCell SSP supports both asynchronous and synchronous operation of the clocks, **PCLK** and **SSPCLK**. Synchronization registers and hand shaking logic have been implemented, and are active at all times. This has a minimal impact on performance or area. Synchronization of control signals is performed on both directions of data flow, that is from the **PCLK** to the **SSPCLK** domain and from the **SSPCLK** to the **PCLK** domain.

2.3 PrimeCell SSP operation

The operation of the PrimeCell SSP is described in the following sections:

- *Interface reset*
- *Configuring the SSP*
- *Enable PrimeCell SSP operation on page 2-8*
- *Clock ratios on page 2-8*
- *Programming the SSPCR0 Control Register on page 2-9*
- *Programming the SSPCR1 Control Register on page 2-9*
- *Frame format on page 2-10*
- *Texas Instruments synchronous serial frame format on page 2-11*
- *Motorola SPI frame format on page 2-12*
- *National Semiconductor Microwire frame format on page 2-18*
- *Examples of master and slave configurations on page 2-21*
- *PrimeCell DMA interface on page 2-25.*

2.3.1 Interface reset

The PrimeCell SSP is reset by the global reset signal **PRESETn** and a block-specific reset signal **nSSPRST**. An external reset controller must use **PRESETn** to assert **nSSPRST** asynchronously and negate it synchronously to **SSPCLK**. **PRESETn** must be asserted LOW for a period long enough to reset the slowest block in the on-chip system, and then taken HIGH again. The PrimeCell SSP requires **PRESETn** to be asserted LOW for at least one period of **PCLK**.

The values of the registers after reset are described in Chapter 3 *Programmer's Model*.

2.3.2 Configuring the SSP

Following reset, the PrimeCell SSP logic is disabled and must be configured when in this state.

Control registers SSPCR0 and SSPCR1 need to be programmed to configure the peripheral as a master or slave operating under one of the following protocols:

- Motorola SPI
- Texas Instruments SSI
- National Semiconductor.

The bit rate, derived from the external **SSPCLK**, requires the programming of the clock prescale register SSPCPSR.

2.3.3 Enable PrimeCell SSP operation

You can either prime the transmit FIFO, by writing up to eight 16-bit values when the PrimeCell SSP is disabled, or allow the transmit FIFO service request to interrupt the CPU. Once enabled, transmission or reception of data begins on the transmit (**SSPTXD**) and receive (**SSPRXD**) pins.

2.3.4 Clock ratios

There is a constraint on the ratio of the frequencies of **PCLK** to **SSPCLK**. The frequency of **SSPCLK** must be less than or equal to that of **PCLK**. This ensures that control signals from the **SSPCLK domain to the PCLK domain are certain to get synchronized before one frame duration**:

$$F_{SSPCLK} \leq F_{PCLK}$$

In the slave mode of operation, the **SSPCLKIN** signal from the external master is double synchronized and then delayed to detect an edge. It takes three **SSPCLKs** to detect an edge on **SSPCLKIN**. **SSPTXD** has less setup time to the falling edge of **SSPCLKIN** on which the master is sampling the line. The setup and hold times on **SSPRXD** with reference to **SSPCLKIN** must be more conservative to ensure that it is at the right value when the actual sampling occurs within the SSPMS. To ensure correct device operation, **SSPCLK** must be at least 12 times faster than the maximum expected frequency of **SSPCLKIN**.

The frequency selected for **SSPCLK** must accommodate the desired range of bit clock rates. The ratio of minimum **SSPCLK** frequency to **SSPCLKOUT** maximum frequency in the case of the slave mode is 12 and for the master mode it is two.

To generate a maximum bit rate of 1.8432Mbps in the Master mode, the frequency of **SSPCLK** must be at least 3.6864MHz. With an **SSPCLK** frequency of 3.6864MHz, the SSPCPSR register has to be programmed with a value of two and the SCR[7:0] field in the SSPCR0 register needs to be programmed as zero.

To work with a maximum bit rate of 1.8432Mbps in the slave mode, the frequency of **SSPCLK** must be at least 22.12MHz. With an **SSPCLK** frequency of 22.12MHz, the SSPCPSR register can be programmed with a value of 12 and the SCR[7:0] field in the SSPCR0 register can be programmed as zero. Similarly the ratio of **SSPCLK** maximum frequency to **SSPCLKOUT** minimum frequency is 254 x 256.

The minimum frequency of **SSPCLK** is governed by the following equations, both of which have to be satisfied:

$$F_{SSPCLK}(\min) \Rightarrow 2 \times F_{SSPCLKOUT}(\max) \text{ [for master mode]}$$

$$F_{SSPCLK}(\min) \Rightarrow 12 \times F_{SSPCLKIN}(\max) \text{ [for slave mode]}$$

The maximum frequency of **SSPCLK** is governed by the following equations, both of which have to be satisfied:

$$F_{\text{SSPCLK}}(\text{max}) \leq 254 \times 256 \times F_{\text{SSPCLKOUT}}(\text{min}) \text{ [for master mode]}$$

$$F_{\text{SSPCLK}}(\text{max}) \leq 254 \times 256 \times F_{\text{SSPCLKIN}}(\text{min}) \text{ [for slave mode]}$$

2.3.5 Programming the SSPCR0 Control Register

Refer to Chapter 3 *Programmer's Model* for more details on the bit assignment of this register.

The SSPCR0 register is used to:

- program the serial clock rate
- select one of the three protocols
- select the data word size (where applicable).

The *Serial Clock Rate* (SCR) value, in conjunction with the SSPCPSR clock prescale divisor value (CPSDVR), is used to derive the PrimeCell SSP transmit and receive bit rate from the external **SSPCLK**.

The frame format is programmed through the FRF bits and the data word size through the DSS bits.

Bit phase and polarity, applicable to Motorola SPI format only, are programmed through the SPH and SPO bits.

2.3.6 Programming the SSPCR1 Control Register

Refer to Chapter 3 *Programmer's Model*, for more details on the bit assignment of this register.

The SSPCR1 register is used to:

- select master or slave mode
- enable a loop back test feature
- enable the PrimeCell SSP peripheral.

To configure the PrimeCell SSP as a master, clear the SSPCR1 register master or slave selection bit (MS) to 0, which is the default value on reset.

Setting the SSPCR1 register MS bit to 1 configures the PrimeCell SSP as a slave. When configured as a slave, enabling or disabling of the PrimeCell SSP **SSPTXD** signal is provided through the SSPCR1 slave mode SSPTXD output disable bit (SOD). This can be used in some multi-slave environments where masters might parallel broadcast.

To enable the operation of the PrimeCell SSP set the *Synchronous Serial Port Enable* (SSE) bit to 1.

Bit rate generation

The serial bit rate is derived by dividing down the input clock **SSPCLK**. The clock is first divided by an even prescale value CPSDVSR from 2 to 254, which is programmed in SSPCPSR. The clock is further divided by a value from 1 to 256, which is $1 + \text{SCR}$, where SCR is the value programmed in SSPCR0.

The frequency of the output signal bit clock **SSPCLKOUT** is defined below:

$$F_{\text{SSPCLKOUT}} = \frac{F_{\text{SSPCLK}}}{\text{CPSDVSR} \times (1 + \text{SCR})}$$

For example, if **SSPCLK** is 3.6864MHz, and CPSDVSR = 2, then **SSPCLKOUT** has a frequency range from 7.2kHz to 1.8432MHz.

2.3.7 Frame format

Each data frame is between 4 and 16 bits long depending on the size of data programmed, and is transmitted starting with the MSB. There are three basic frame types that can be selected:

- Texas Instruments synchronous serial
- Motorola SPI
- National Semiconductor Microwire.

For all three formats, the serial clock (**SSPCLKOUT**) is held inactive while the PrimeCell SSP is idle, and transitions at the programmed frequency only during active transmission or reception of data. The idle state of **SSPCLKOUT** is utilized to provide a receive timeout indication that occurs when the receive FIFO still contains data after a timeout period.

For Motorola SPI and National Semiconductor Microwire frame formats, the serial frame (**SSPFSSOUT**) pin is active LOW, and is asserted (pulled down) during the entire transmission of the frame.

For Texas Instruments synchronous serial frame format, the **SSPFSSOUT** pin is pulsed for one serial clock period starting at its rising edge, prior to the transmission of each frame. For this frame format, both the PrimeCell SSP and the off-chip slave device drive their output data on the rising edge of **SSPCLKOUT**, and latch data from the other device on the falling edge.

Unlike the full-duplex transmission of the other two frame formats, the National Semiconductor Microwire format uses a special master-slave messaging technique, which operates at half-duplex. In this mode, when a frame begins, an 8-bit control message is transmitted to the off-chip slave. During this transmit, no incoming data is received by the SSP. After the message has been sent, the off-chip slave decodes it and, after waiting one serial clock after the last bit of the 8-bit control message has been sent, responds with the requested data. The returned data can be 4 to 16 bits in length, making the total frame length anywhere from 13 to 25 bits.

2.3.8 Texas Instruments synchronous serial frame format

Figure 2-2 shows the Texas Instruments synchronous serial frame format for a single transmitted frame.

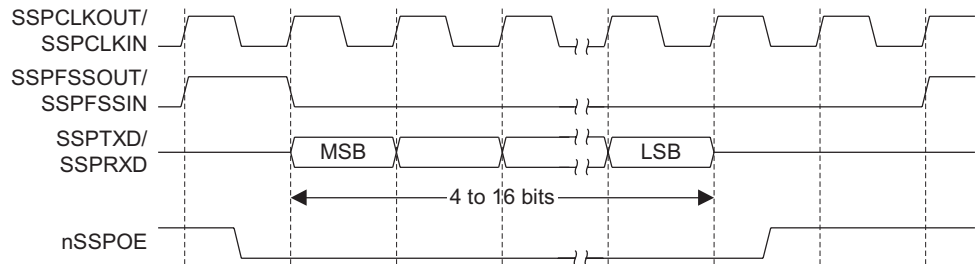


Figure 2-2 Texas Instruments synchronous serial frame format (single transfer)

In this mode, **SSPCLKOUT** and **SSPFSSOUT** are forced LOW, and the transmit data line **SSPTXD** is tristated whenever the PrimeCell SSP is idle. Once the bottom entry of the transmit FIFO contains data, **SSPFSSOUT** is pulsed HIGH for one **SSPCLKOUT** period. The value to be transmitted is also transferred from the transmit FIFO to the serial shift register of the transmit logic. On the next rising edge of **SSPCLKOUT**, the MSB of the 4 to 16-bit data frame is shifted out on the **SSPTXD** pin. Likewise, the MSB of the received data is shifted onto the **SSPRXD** pin by the off-chip serial slave device.

Both the PrimeCell SSP and the off-chip serial slave device then clock each data bit into their serial shifter on the falling edge of each **SSPCLKOUT**. The received data is transferred from the serial shifter to the receive FIFO on the first rising edge of **SSPCLKOUT** after the LSB has been latched.

Figure 2-3 on page 2-12 shows the Texas Instruments synchronous serial frame format when back-to-back frames are transmitted.

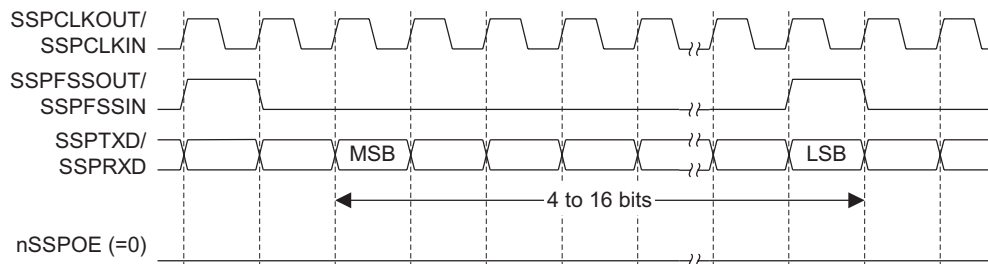


Figure 2-3 TI synchronous serial frame format (continuous transfer)

2.3.9 Motorola SPI frame format

The Motorola SPI interface is a four-wire interface where the **SSPFSSOUT** signal behaves as a slave select. The main feature of the Motorola SPI format is that the inactive state and phase of the **SSPCLKOUT** signal are programmable through the SPO and SPH bits within the SSPSCR0 control register.

SSPCLKOUT, clock polarity

When the SPO clock polarity control bit is LOW, it produces a steady state low value on the **SSPCLKOUT** pin. If the SPO clock polarity control bit is HIGH, a steady state high value is placed on the **SSPCLKOUT** pin when data is not being transferred.

SSPCLKOUT, clock phase

The SPH control bit selects the clock edge that captures data and allows it to change state. It has the most impact on the first bit transmitted by either allowing or not allowing a clock transition before the first data capture edge.

When the SPH phase control bit is LOW, data is captured on the first clock edge transition. If the SPH clock phase control bit is HIGH, data is captured on the second clock edge transition.

2.3.10 Motorola SPI Format with SPO=0, SPH=0

Single and continuous transmission signal sequences for Motorola SPI format with SPO=0, SPH=0 are shown in Figure 2-4 on page 2-13 and Figure 2-5 on page 2-13.

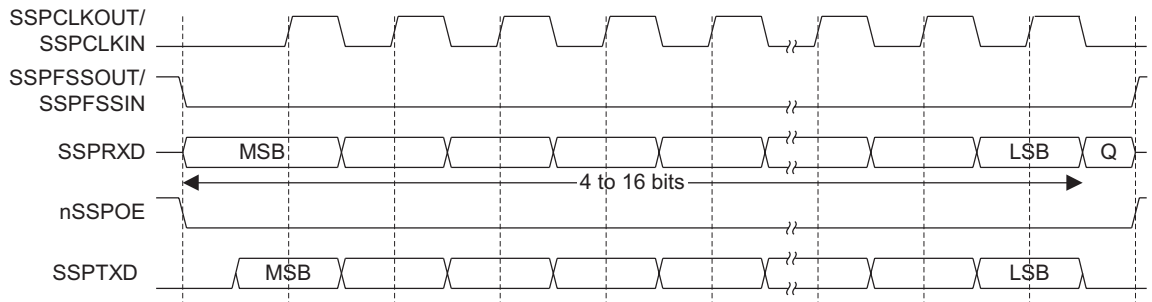


Figure 2-4 Motorola SPI frame format (single transfer) with SPO=0 and SPH=0

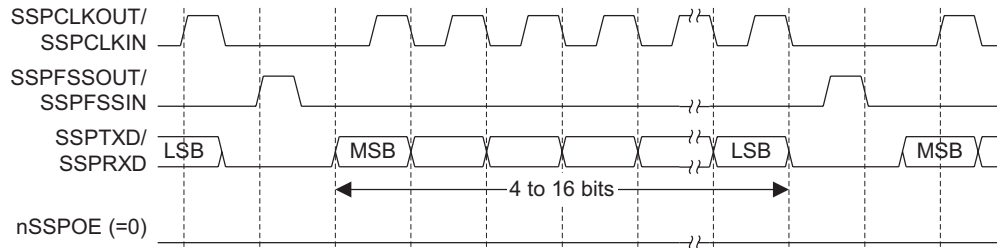


Figure 2-5 Motorola SPI frame format (continuous transfer) with SPO=0 and SPH=0

In this configuration, during idle periods:

- the **SSPCLKOUT** signal is forced LOW
- **SSPFSSOUT** is forced HIGH
- the transmit data line **SSPTXD** is arbitrarily forced LOW
- the **nSSPOE** pad enable signal is forced HIGH, making the transmit pad high impedance
- when the PrimeCell SSP is configured as a master, the **nSSPCTLOE** line is driven LOW, enabling the **SSPCLKOUT** pad (active LOW enable)
- when the PrimeCell SSP is configured as a slave, the **nSSPCTLOE** line is driven HIGH, disabling the **SSPCLKOUT** pad (active LOW enable).

If the PrimeCell SSP is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the **SSPFSSOUT** master signal being driven LOW. This causes slave data to be enabled onto the **SSPRXD** input line of the master. The **nSSPOE** line is driven LOW, enabling the master **SSPTXD** output pad.

One half **SSPCLKOUT** period later, valid master data is transferred to the **SSPTXD** pin. Now that both the master and slave data have been set, the **SSPCLKOUT** master clock pin goes HIGH after one further half **SSPCLKOUT** period.

The data is now captured on the rising and propagated on the falling edges of the **SSPCLKOUT** signal.

In the case of a single word transmission, after all bits of the data word have been transferred, the **SSPFSSOUT** line is returned to its idle HIGH state one **SSPCLKOUT** period after the last bit has been captured.

However, in the case of continuous back-to-back transmissions, the **SSPFSSOUT** signal must be pulsed HIGH between each data word transfer. This is because the slave select pin freezes the data in its serial peripheral register and does not allow it to be altered if the SPH bit is logic zero. Therefore the master device must raise the **SSPFSSIN** pin of the slave device between each data transfer to enable the serial peripheral data write. On completion of the continuous transfer, the **SSPFSSOUT** pin is returned to its idle state one **SSPCLKOUT** period after the last bit has been captured.

2.3.11 Motorola SPI Format with SPO=0, SPH=1

The transfer signal sequence for Motorola SPI format with SPO=0, SPH=1 is shown in Figure 2-6, which covers both single and continuous transfers.

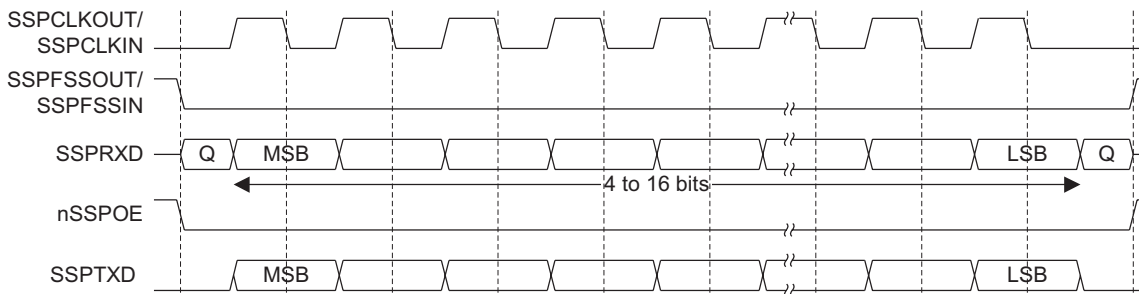


Figure 2-6 Motorola SPI frame format with SPO=0 and SPH=1

In this configuration, during idle periods:

- the **SSPCLKOUT** signal is forced LOW
- **SSPFSSOUT** is forced HIGH
- the transmit data line **SSPTXD** is arbitrarily forced LOW
- the **nSSPOE** pad enable signal is forced HIGH, making the transmit pad high impedance

- when the PrimeCell SSP is configured as a master, the **nSSPCTL0E** line is driven LOW, enabling the **SSPCLKOUT** pad (active LOW enable)
- when the PrimeCell SSP is configured as a slave, the **nSSPCTL0E** line is driven HIGH, disabling the **SSPCLKOUT** pad (active LOW enable).

If the PrimeCell SSP is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the **SSPFSSOUT** master signal being driven LOW. The **nSSPOE** line is driven LOW, enabling the master **SSPTXD** output pad. After a further one half **SSPCLKOUT** period, both master and slave valid data is enabled onto their respective transmission lines. At the same time, the **SSPCLKOUT** is enabled with a rising edge transition.

Data is then captured on the falling edges and propagated on the rising edges of the **SSPCLKOUT** signal.

In the case of a single word transfer, after all bits have been transferred, the **SSPFSSOUT** line is returned to its idle HIGH state one **SSPCLKOUT** period after the last bit has been captured.

For continuous back-to-back transfers, the **SSPFSSOUT** pin is held LOW between successive data words and termination is the same as that of the single word transfer.

2.3.12 Motorola SPI Format with SPO=1, SPH=0

Single and continuous transmission signal sequences for Motorola SPI format with SPO=1, SPH=0 are shown in Figure 2-7 and Figure 2-8 on page 2-16.

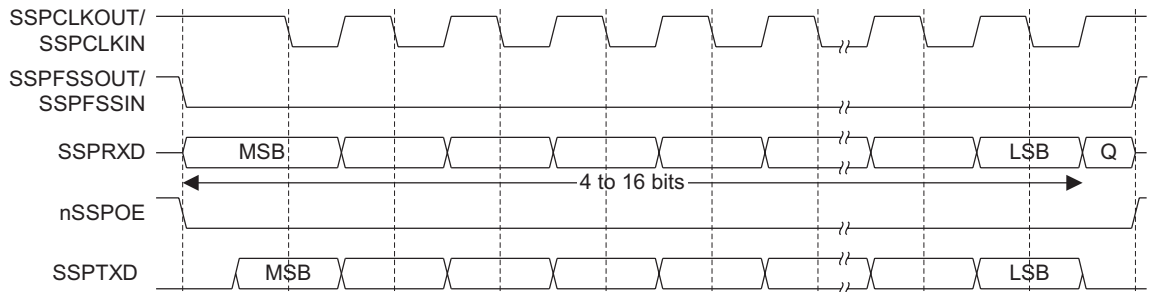


Figure 2-7 Motorola SPI frame format (single transfer) with SPO=1 and SPH=0

———— Note ————

In Figure 2-7, Q is an undefined signal.

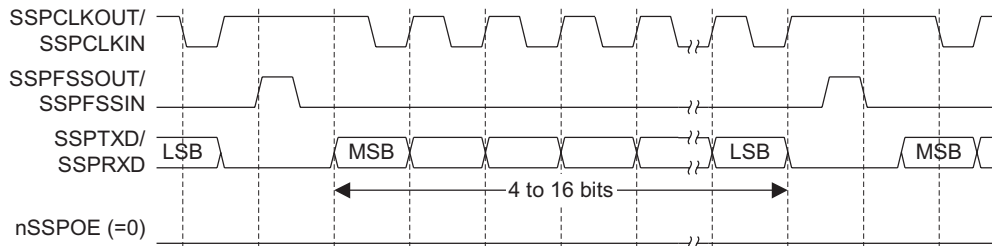


Figure 2-8 Motorola SPI frame format (continuous transfer) with SPO=1 and SPH=0

In this configuration, during idle periods

- the **SSPCLKOUT** signal is forced HIGH
- **SSPFSSOUT** is forced HIGH
- the transmit data line **SSPTXD** is arbitrarily forced LOW
- the **nSSPOE** pad enable signal is forced HIGH, making the transmit pad high impedance
- when the PrimeCell SSP is configured as a master, the **nSSPCTL0E** line is driven LOW, enabling the **SSPCLKOUT** pad (active LOW enable)
- when the PrimeCell SSP is configured as a slave, the **nSSPCTL0E** line is driven HIGH, disabling the **SSPCLKOUT** pad (active LOW enable).

If the PrimeCell SSP is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the **SSPFSSOUT** master signal being driven LOW, which causes slave data to be immediately transferred onto the **SSPRXD** line of the master. The **nSSPOE** line is driven LOW, enabling the master **SSPTXD** output pad.

One half period later, valid master data is transferred to the **SSPTXD** line. Now that both the master and slave data have been set, the **SSPCLKOUT** master clock pin becomes LOW after one further half **SSPCLKOUT** period. This means that data is captured on the falling edges and be propagated on the rising edges of the **SSPCLKOUT** signal.

In the case of a single word transmission, after all bits of the data word are transferred, the **SSPFSSOUT** line is returned to its idle HIGH state one **SSPCLKOUT** period after the last bit has been captured.

However, in the case of continuous back-to-back transmissions, the **SSPFSSOUT** signal must be pulsed HIGH between each data word transfer. This is because the slave select pin freezes the data in its serial peripheral register and does not allow it to be altered if the SPH bit is logic zero. Therefore the master device must raise the

SSPFSSIN pin of the slave device between each data transfer to enable the serial peripheral data write. On completion of the continuous transfer, the **SSPFSSOUT** pin is returned to its idle state one **SSPCLKOUT** period after the last bit has been captured.

2.3.13 Motorola SPI Format with SPO=1, SPH=1

The transfer signal sequence for Motorola SPI format with SPO=1, SPH=1 is shown in Figure 2-9, which covers both single and continuous transfers.

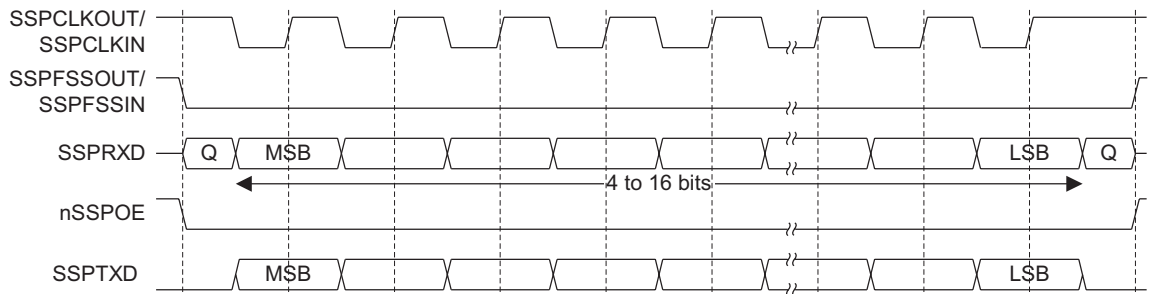


Figure 2-9 Motorola SPI frame format with SPO=1 and SPH=1

Note

In Figure 2-9, Q is an undefined signal.

In this configuration, during idle periods:

- the **SSPCLKOUT** signal is forced HIGH
- **SSPFSSOUT** is forced HIGH
- the transmit data line **SSPTXD** is arbitrarily forced LOW
- the **nSSPOE** pad enable signal is forced HIGH, making the transmit pad high impedance
- when the PrimeCell SSP is configured as a master, the **nSSPCTLOE** line is driven LOW, enabling the **SSPCLKOUT** pad (active LOW enable)
- when the PrimeCell SSP is configured as a slave, the **nSSPCTLOE** line is driven HIGH, disabling the **SSPCLKOUT** pad (active LOW enable).

If the PrimeCell SSP is enabled and there is valid data within the transmit FIFO, the start of transmission is signified by the **SSPFSSOUT** master signal being driven LOW. The **nSSPOE** line is driven LOW, enabling the master **SSPTXD** output pad. After a further one half **SSPCLKOUT** period, both master and slave data are enabled onto their

respective transmission lines. At the same time, the **SSPCLKOUT** is enabled with a falling edge transition. Data is then captured on the rising edges and propagated on the falling edges of the **SSPCLKOUT** signal.

After all bits have been transferred, in the case of a single word transmission, the **SSPFSSOUT** line is returned to its idle HIGH state one **SSPCLKOUT** period after the last bit has been captured.

For continuous back-to-back transmissions, the **SSPFSSOUT** pins remains in its active LOW state, until the final bit of the last word has been captured, and then returns to its idle state as described above.

For continuous back-to-back transfers, the **SSPFSSOUT** pin is held LOW between successive data words and termination is the same as that of the single word transfer.

2.3.14 National Semiconductor Microwire frame format

Figure 2-10 shows the National Semiconductor Microwire frame format, again for a single frame. Figure 2-11 on page 2-20 shows the same format when back to back frames are transmitted.

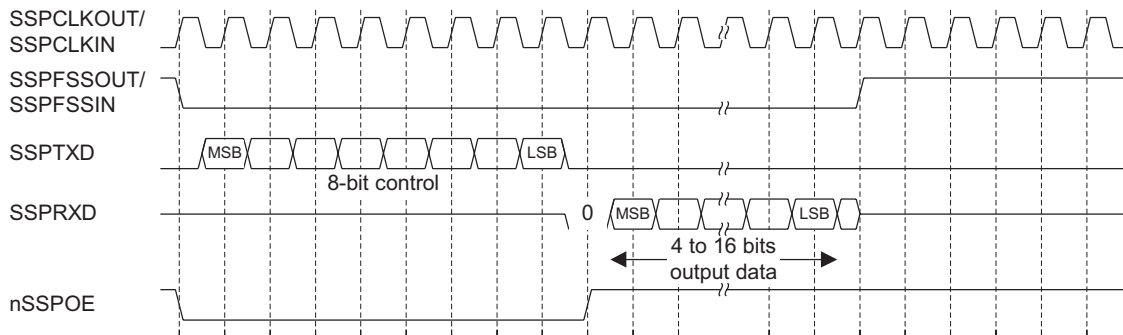


Figure 2-10 Microwire frame format (single transfer)

Microwire format is very similar to SPI format, except that transmission is half-duplex instead of full-duplex, using a master-slave message passing technique. Each serial transmission begins with an 8-bit control word that is transmitted from the PrimeCell SSP to the off-chip slave device. During this transmission, no incoming data is received by the PrimeCell SSP. After the message has been sent, the off-chip slave decodes it and, after waiting one serial clock after the last bit of the 8-bit control message has been sent, responds with the required data. The returned data is 4 to 16 bits in length, making the total frame length anywhere from 13 to 25 bits.

In this configuration, during idle periods:

- the **SSPCLKOUT** signal is forced LOW
- **SSPFSSOUT** is forced HIGH
- the transmit data line **SSPTXD** is arbitrarily forced LOW
- the **nSSPOE** pad enable signal is forced HIGH, making the transmit pad high impedance.

A transmission is triggered by writing a control byte to the transmit FIFO. The falling edge of **SSPFSSOUT** causes the value contained in the bottom entry of the transmit FIFO to be transferred to the serial shift register of the transmit logic, and the MSB of the 8-bit control frame to be shifted out onto the **SSPTXD** pin. **SSPFSSOUT** remains LOW for the duration of the frame transmission. The **SSPRXD** pin remains tristated during this transmission.

The off-chip serial slave device latches each control bit into its serial shifter on the rising edge of each **SSPCLKOUT**. After the last bit is latched by the slave device, the control byte is decoded during a one clock wait-state, and the slave responds by transmitting data back to the PrimeCell SSP. Each bit is driven onto **SSPRXD** line on the falling edge of **SSPCLKOUT**. The PrimeCell SSP in turn latches each bit on the rising edge of **SSPCLKOUT**. At the end of the frame, for single transfers, the **SSPFSSOUT** signal is pulled HIGH one clock period after the last bit has been latched in the receive serial shifter, that causes the data to be transferred to the receive FIFO.

————— **Note** —————

The off-chip slave device can tristate the receive line either on the falling edge of **SSPCLKOUT** after the LSB has been latched by the receive shifter, or when the **SSPFSSOUT** pin goes HIGH.

For continuous transfers, data transmission begins and ends in the same manner as a single transfer. However, the **SSPFSSOUT** line is continuously asserted (held LOW) and transmission of data occurs back to back. The control byte of the next frame follows directly after the LSB of the received data from the current frame. Each of the received values is transferred from the receive shifter on the falling edge **SSPCLKOUT**, after the LSB of the frame has been latched into the PrimeCell SSP.

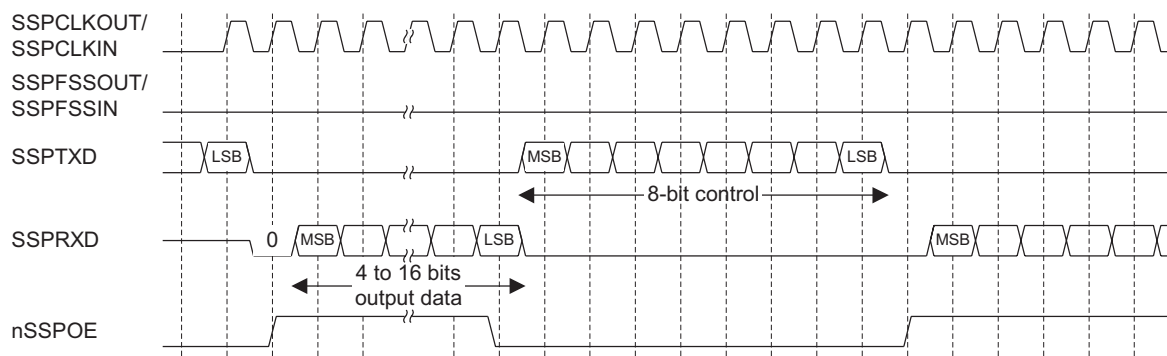


Figure 2-11 Microwire frame format (continuous transfers)

Setup and hold time requirements on SSPFSSIN with respect to SSPCLKIN in Microwire mode

In the Microwire mode, the PrimeCell SSP slave samples the first bit of receive data on the rising edge of **SSPCLKIN** after **SSPFSSIN** has gone LOW. Masters that drive a free-running **SSPCKIN** must ensure that the **SSPFSSIN** signal has sufficient setup and hold margins with respect to the rising edge of **SSPCLKIN**.

Figure 2-12 on page 2-21 illustrates these setup and hold time requirements. With respect to the **SSPCLKIN** rising edge on which the first bit of receive data is to be sampled by the PrimeCell SSP slave, **SSPFSSIN** must have a setup of at least two times the period of **SSPCLKIN** on which the PrimeCell SSP operates. With respect to the **SSPCLKIN** rising edge previous to this edge, **SSPFSSIN** must have a hold of at least one **SSPCLKIN** period.

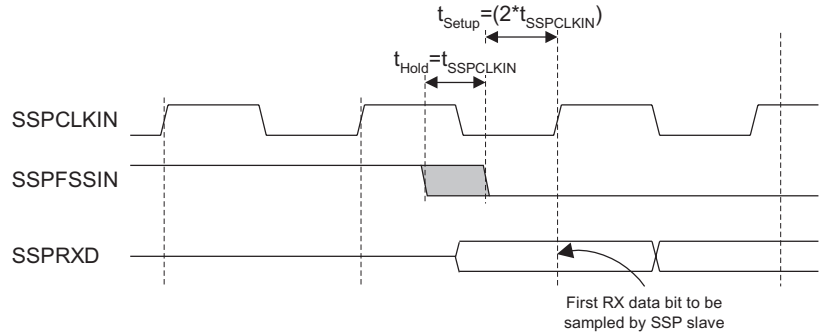


Figure 2-12 Microwire frame format, SSPFSSIN input setup and hold requirements

2.3.15 Examples of master and slave configurations

Figure 2-13 on page 2-22, Figure 2-14 on page 2-23 and Figure 2-15 on page 2-24 show how the PrimeCell SSP (PL022) peripheral can be connected to other synchronous serial peripherals, when it is configured as a master or slave.

Note

The SSP (PL022) does not support dynamic switching between master and slave in a system. Each instance is configured and connected either as a master or slave.

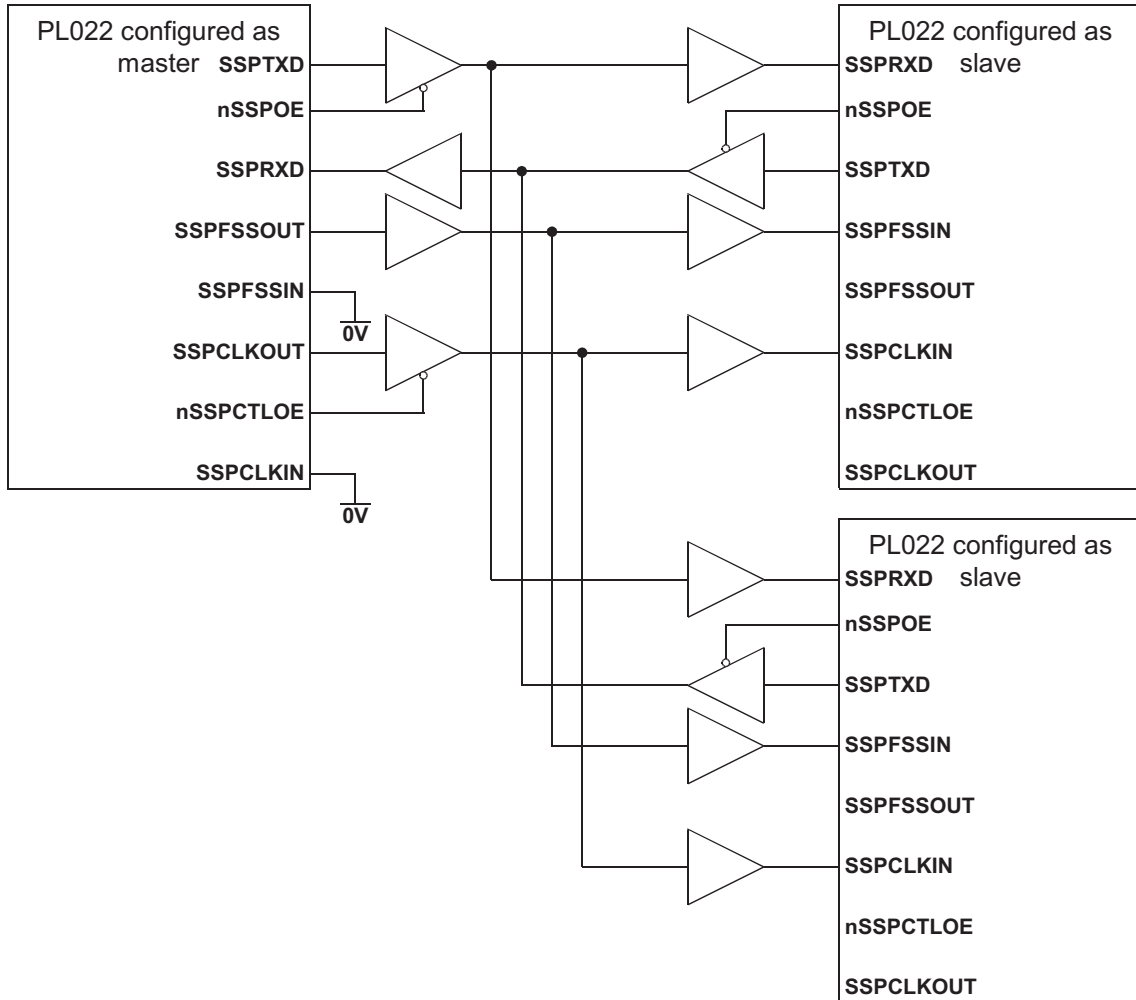


Figure 2-13 PrimeCell SSP master coupled to two slaves

Figure 2-13 shows the PrimeCell SSP (PL022) instantiated three times, as a single master and two slaves. The master can broadcast to the two slaves through the master **SSPTXD** line. In response, only one slave drives its **nSSPOE** signal HIGH, thereby enabling its **SSPTXD** data onto the **SSPRXD** line of the master.

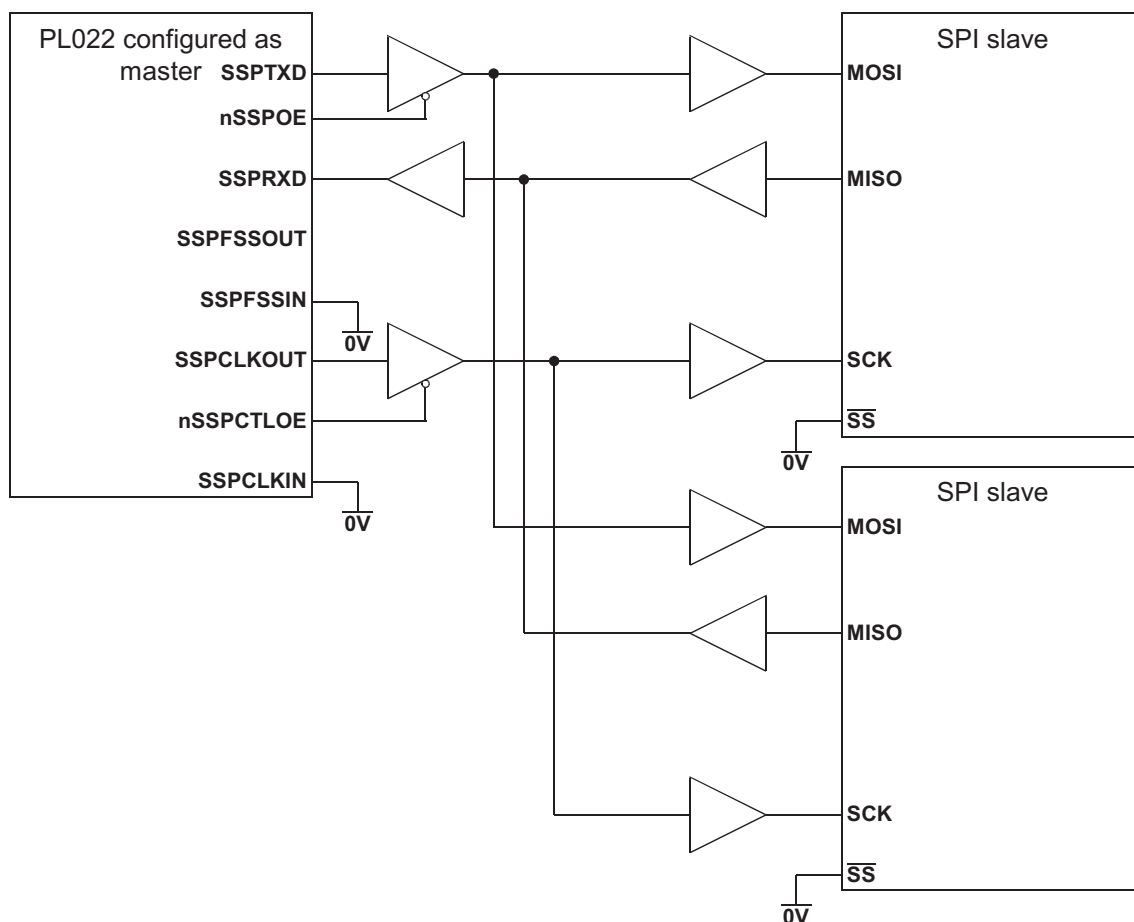


Figure 2-14 PrimeCell SSP master coupled to two SPI slaves

Figure 2-14 shows how an PrimeCell SSP (PL022), configured as master, interfaces to two Motorola SPI slaves. Each SPI *Slave Select* (SS) signal is permanently tied LOW and configures them as slaves. Similar to the above operation, the master can broadcast to the two slaves through the master PrimeCell SSP **SSPTXD** line. In response, only one slave drives its SPI MISO port onto the **SSPRXD** line of the master.

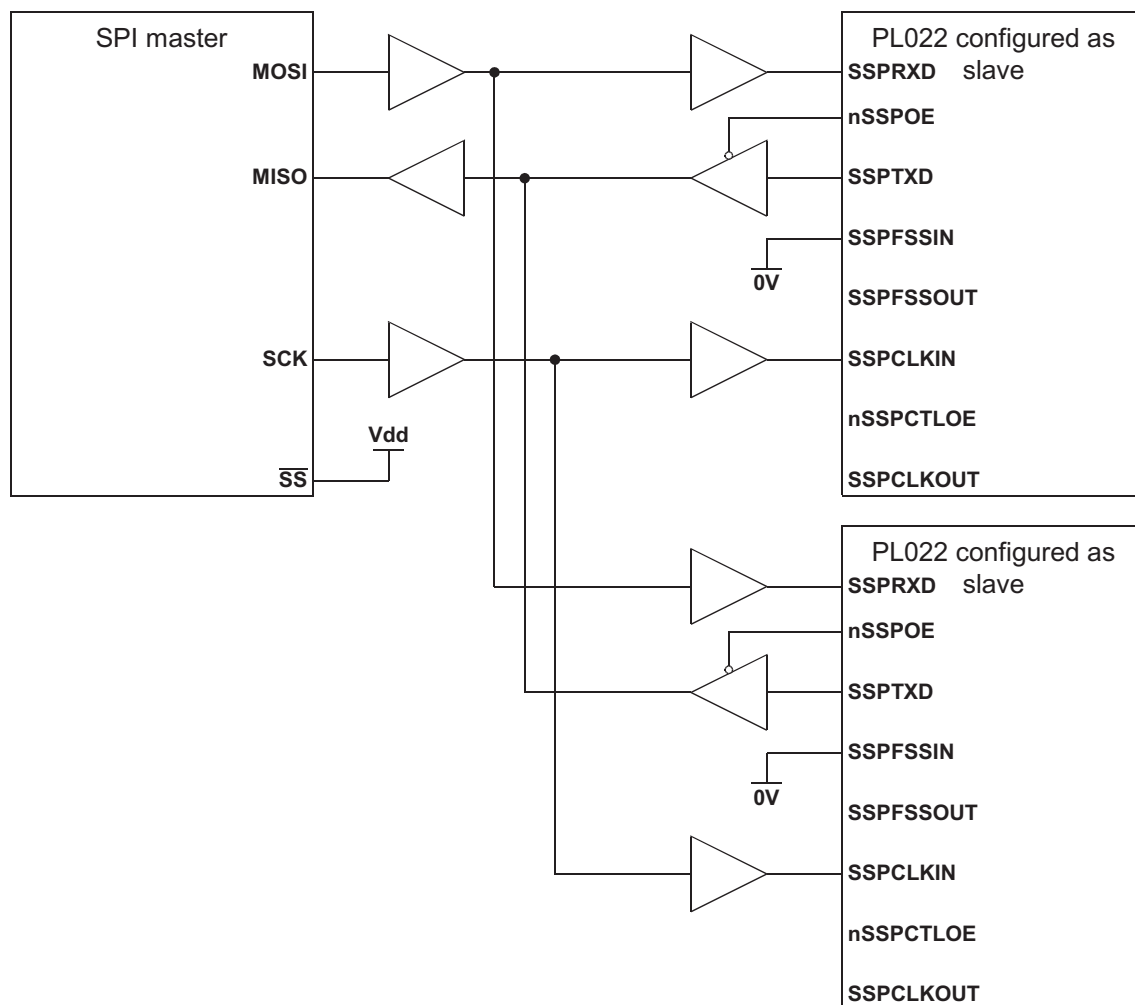


Figure 2-15 SPI master coupled to two PrimeCell SSP slaves

Figure 2-15 shows a Motorola SPI configured as a master and interfaced to two instances of PrimeCell SSP (PL022) configured as slaves. In this case the slave *Select Signal* (SS) is permanently tied HIGH and configures it as a master. The master can broadcast to the two slaves through the master SPI MOSI line and in response, only one slave drives its **nSSPOE** signal LOW. This enables its **SSPTXD** data onto the MISO line of the master.

2.3.16 PrimeCell DMA interface

The PrimeCell SSP provides an interface to connect to the DMA controller. The DMA operation of the PrimeCell SSP is controlled through the PrimeCell SSP DMA control register, SSPDMACR. The DMA interface includes the following signals, for receive:

SSPRXDMASREQ

Single-character DMA transfer request, asserted by the SSP. This signal is asserted when the receive FIFO contains at least one character.

SSPRXDMABREQ

Burst DMA transfer request, asserted by the SSP. This signal is asserted when the receive FIFO contains four or more characters.

SSPRXDMACLR

DMA request clear, asserted by the DMA controller to clear the receive request signals. If DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.

The DMA interface includes the following signals, for transmit:

SSPTXDMASREQ

Single-character DMA transfer request, asserted by the SSP. This signal is asserted when there is at least one empty location in the transmit FIFO.

SSPTXDMABREQ

Burst DMA transfer request, asserted by the SSP. This signal is asserted when the transmit FIFO contains four or less characters.

SSPTXDMACLR

DMA request clear, asserted by the DMA controller to clear the transmit request signals. If DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.

The burst transfer and single transfer request signals are not mutually exclusive. They can both be asserted at the same time. For example, when there is more data than the watermark level of four in the receive FIFO, the burst transfer request and the single transfer request are asserted. When the amount of data left in the receive FIFO is less than the watermark level, the single request only is asserted. This is useful for situations where the number of characters left to be received in the stream is less than a burst.

For example, say 19 characters have to be received. The DMA controller then transfers four bursts of four characters and three single transfers to complete the stream.

Note

For the remaining three characters the PrimeCell SSP does not assert the burst request.

Each request signal remains asserted until the relevant DMA clear signal is asserted. After the request clear signal is deasserted, a request signal can become active again, depending on the conditions described above. All request signals are deasserted if the PrimeCell SSP is disabled or the DMA enable signal is cleared.

Table 2-1 shows the trigger points for **DMABREQ**, for both the transmit and receive FIFOs.

Table 2-1 DMA trigger points for the transmit and receive FIFOs

Watermark level	Burst length	
	Transmit (number of empty locations)	Receive (number of filled locations)
1/2	4	4

Figure 2-16 shows the timing diagram for both a single transfer request and a burst transfer request with the appropriate DMA clear signal. The signals are all synchronous to **PCLK**.

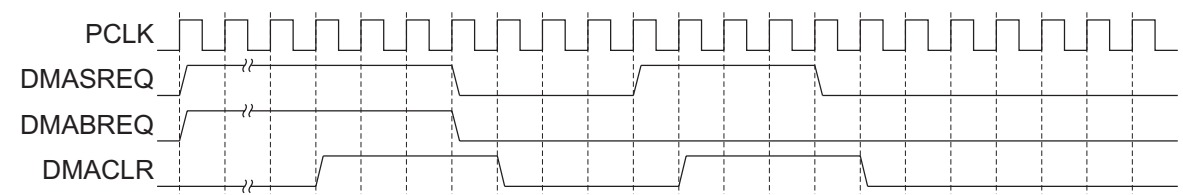


Figure 2-16 DMA transfer waveforms

Chapter 3

Programmer's Model

This chapter describes the ARM PrimeCell Synchronous Serial Port (PL022) registers and provides details needed when programming the microcontroller. It contains the following sections:

- *About the programmer's model* on page 3-2
- *Summary of PrimeCell SSP registers* on page 3-3
- *Register descriptions* on page 3-5
- *Interrupts* on page 3-19.

3.1 About the programmer's model

The base address of the PrimeCell SSP is not fixed, and might be different for any particular system implementation. The offset of any particular register from the base address, however, is fixed.

The following locations are reserved, and must not be used during normal operation:

- locations at offsets +0x028 to +0x07C and +0xFD0 to +0xFDC are reserved for possible future extensions
- locations at offsets +0x080 to +0x088 are reserved for test purposes.

3.2 Summary of PrimeCell SSP registers

The PrimeCell SSP registers are shown in Table 3-1.

Table 3-1 PrimeCell SSP register summary

Address	Type	Width	Reset value	Name	Description
SSP Base + 0x00	Read/ write	16	0x0000	SSPCR0	Control register 0. See Table 3-2 on page 3-5.
SSP Base + 0x04	Read/ write	4	0x0	SSPCR1	Control register 1. See Table 3-3 on page 3-7.
SSP Base + 0x08	Read/ write	16	0x----	SSPDR	Receive FIFO (read) and transmit FIFO data register (write). See Table 3-4 on page 3-8.
SSP Base + 0x0C	Read	5	0x03	SSPSR	Status register. See Table 3-5 on page 3-9.
SSP Base + 0x10	Read/ write	8	0x00	SSPCPSR	Clock prescale register. See Table 3-6 on page 3-10.
SSP Base + 0x14	Read/ write	4	0x0	SSPIMSC	Interrupt mask set and clear register. See Table 3-7 on page 3-11.
SSP Base + 0x18	Read	4	0x8	SSPRIS	Raw interrupt status register. See Table 3-8 on page 3-12.
SSP Base + 0x1C	Read	4	0x0	SSPMIS	Masked interrupt status register. See Table 3-9 on page 3-12.
SSP Base + 0x20	Write	4	0x0	SSPICR	Interrupt clear register. See Table 3-10 on page 3-13.
SSP Base + 0x24	Read/ write	2	0x0	SSPDMACR	DMA control register. See Table 3-11 on page 3-13.
SSP Base + 0x28 to 0x7C	-	-	-	-	Reserved
SSP Base + 0x80 to 0x8C	-	-	-	-	Reserved for test
SSP Base + 0x90 to 0xFCC	-	-	-	-	Reserved

Table 3-1 PrimeCell SSP register summary (continued)

Address	Type	Width	Reset value	Name	Description
SSP Base + 0xFD0 to 0xFDC	-	-	-	-	Reserved for future expansion
SSP base + 0xFE0	Read	8	0x22	SSPPeriphID0	Peripheral identification register bits 7:0. See Table 3-12 on page 3-15.
SSP base + 0xFE4	Read	8	0x10	SSPPeriphID1	Peripheral identification register bits 15:8. See Table 3-13 on page 3-15.
SSP base + 0xFE8	Read	8	0x04	SSPPeriphID2	Peripheral identification register bits 23:16. See Table 3-14 on page 3-16.
SSP base + 0xFEC	Read	8	0x00	SSPPeriphID3	Peripheral identification register bits 31:24. See Table 3-15 on page 3-16.
SSP base + 0xFF0	Read	8	0x0D	SSPPCellID0	PrimeCell identification register bits 7:0. See Table 3-16 on page 3-17.
SSP base + 0xFF4	Read	8	0xF0	SSPPCellID1	PrimeCell identification register bits 15:8. See Table 3-17 on page 3-18.
SSP base + 0xFF8	Read	8	0x05	SSPPCellID2	PrimeCell identification register bits 23:16. See Table 3-18 on page 3-18.
SSP base + 0xFFC	Read	8	0xB1	SSPPCellID3	PrimeCell identification register bits 31:24. See Table 3-19 on page 3-18.

3.3 Register descriptions

The following registers are described in this section:

- *Control register 0, SSPCR0*
- *Control register 1, SSPCR1* on page 3-7
- *Data register, SSPDR* on page 3-8
- *Status register, SSPSR* on page 3-9
- *Clock prescale register, SSPCPSR* on page 3-10
- *Interrupt mask set or clear register, SSPIMSC* on page 3-11
- *Raw interrupt status register, SSPRIS* on page 3-12
- *Masked interrupt status register, SSPMIS* on page 3-12
- *Interrupt clear register, SSPICR* on page 3-13
- *DMA control register, SSPDMACR* on page 3-13
- *Peripheral identification registers, SSPPeriphID0-3* on page 3-14
- *PrimeCell identification registers, SSPPCellID0-3* on page 3-16.

3.3.1 Control register 0, SSPCR0

SSPCR0 is control register 0 and contains five bit fields that control various functions within the PrimeCell SSP. Table 3-2 shows the bit assignments for SSPCR0.

Table 3-2 SSPCR0 register

Bits	Name	Type	Function
15:8	SCR	Read/write	Serial clock rate. The value SCR is used to generate the transmit and receive bit rate of the PrimeCell SSP. The bit rate is: $\frac{F_{SSPCLK}}{CPSDVR \times (1+SCR)}$ where CPSDVR is an even value from 2 to 254, programmed through the SSPCPSR register and SCR is a value from 0 to 255.
7	SPH	Read/write	SSPCLKOUT phase (applicable to Motorola SPI frame format only). See <i>Motorola SPI frame format</i> on page 2-12.

Table 3-2 SSPCR0 register (continued)

Bits	Name	Type	Function
6	SPO	Read/write	SSPCLKOUT polarity (applicable to Motorola SPI frame format only). See <i>Motorola SPI frame format</i> on page 2-12.
5:4	FRF	Read/write	Frame format: 00 = Motorola SPI frame format 01 = TI synchronous serial frame format 10 = National Microwire frame format 11 = Reserved, undefined operation
3:0	DSS	Read/write	Data Size Select: 0000 = Reserved, undefined operation 0001 = Reserved, undefined operation 0010 = Reserved, undefined operation 0011 = 4-bit data 0100 = 5-bit data 0101 = 6-bit data 0110 = 7-bit data 0111 = 8-bit data 1000 = 9-bit data 1001 = 10-bit data 1010 = 11-bit data 1011 = 12-bit data 1100 = 13-bit data 1101 = 14-bit data 1110 = 15-bit data 1111 = 16-bit data.

3.3.2 Control register 1, SSPCR1

SSPCR1 is the control register 1 and contains four different bit fields, which control various functions within the PrimeCell SSP. Table 3-3 shows the bit assignments for SSPCR1.

Table 3-3 SSPCR1 register

Bits	Name	Type	Function
15:4	-	-	Reserved, read unpredictable, should be written as 0.
3	SOD	Read/write	Slave-mode output disable. This bit is relevant only in the slave mode (MS=1). In multiple-slave systems, it is possible for an PrimeCell SSP master to broadcast a message to all slaves in the system while ensuring that only one slave drives data onto its serial output line. In such systems the RXD lines from multiple slaves could be tied together. To operate in such systems, the SOD bit can be set if the PrimeCell SSP slave is not supposed to drive the SSPTXD line. 0 = SSP can drive the SSPTXD output in slave mode. 1 = SSP must not drive the SSPTXD output in slave mode.
2	MS	Read/write	Master or slave mode select. This bit can be modified only when the PrimeCell SSP is disabled (SSE=0): 0 = device configured as master (default) 1 = device configured as slave.
1	SSE	Read/write	Synchronous serial port enable: 0 = SSP operation disabled 1 = SSP operation enabled.
0	LBM	Read/write	Loop back mode: 0 = Normal serial port operation enabled 1 = Output of transmit serial shifter is connected to input of receive serial shifter internally.

3.3.3 Data register, SSPDR

SSPDR is the data register and is 16-bits wide. When SSPDR is read, the entry in the receive FIFO (pointed to by the current FIFO read pointer) is accessed. As data values are removed by the PrimeCell SSP receive logic from the incoming data frame, they are placed into the entry in the receive FIFO (pointed to by the current FIFO write pointer).

When SSPDR is written to, the entry in the transmit FIFO (pointed to by the write pointer), is written to. Data values are removed from the transmit FIFO one value at a time by the transmit logic. It is loaded into the transmit serial shifter, then serially shifted out onto the **SSPTXD** pin at the programmed bit rate.

When a data size of less than 16 bits is selected, the user must right-justify data written to the transmit FIFO. The transmit logic ignores the unused bits. Received data less than 16 bits is automatically right-justified in the receive buffer.

When the PrimeCell SSP is programmed for National Microwire frame format, the default size for transmit data is eight bits (the most significant byte is ignored). The receive data size is controlled by the programmer. The transmit FIFO and the receive FIFO are not cleared even when SSE is set to zero. This allows the software to fill the transmit FIFO before enabling the PrimeCell SSP. Table 3-4 shows the bit assignments for SSPDR.

Table 3-4 SSPDR register

Bits	Name	Type	Function
15:0	DATA	Read/write	Transmit/Receive FIFO: Read = Receive FIFO Write = Transmit FIFO. You must right-justify data when the PrimeCell SSP is programmed for a data size that is less than 16 bits. Unused bits at the top are ignored by transmit logic. The receive logic automatically right-justifies.

3.3.4 Status register, SSPSR

SSPSR is a read-only status register that contains bits that indicate the FIFO fill status and the PrimeCell SSP busy status. Table 3-5 shows the bit assignments for SSPSR.

Table 3-5 SSPSR register

Bits	Name	Type	Function
15:5	-	-	Reserved, read unpredictable, should be written as 0.
4	BSY	Read	PrimeCell SSP busy flag (read-only): 0 = SSP is idle 1 = SSP is currently transmitting and/or receiving a frame or the transmit FIFO is not empty.
3	RFF	Read	Receive FIFO full (read-only): 0 = Receive FIFO is not full 1 = Receive FIFO is full.
2	RNE	Read	Receive FIFO not empty (read-only): 0 = Receive FIFO is empty 1 = Receive FIFO is not empty.
1	TNF	Read	Transmit FIFO not full (read-only): 0 = Transmit FIFO is full 1 = Transmit FIFO is not full.
0	TFE	Read	Transmit FIFO empty (read-only): 0 = Transmit FIFO is not empty 1 = Transmit FIFO is empty.

3.3.5 Clock prescale register, SSPCPSR

SSPCPSR is the clock prescale register and specifies the division factor by which the input **SSPCLK** must be internally divided before further use.

The value programmed into this register must be an even number between 2 to 254. The least significant bit of the programmed number is hard-coded to zero. If an odd number is written to this register, data read back from this register has the least significant bit as zero. Table 3-6 shows the bit assignments for SSPCPSR.

Table 3-6 SSPCPSR register

Bits	Name	Type	Function
15:8	-	-	Reserved, read unpredictable, must be written as 0.
7:0	CPSDVSR	Read/write	Clock prescale divisor. Must be an even number from 2 to 254, depending on the frequency of SSPCLK . The least significant bit always returns zero on reads.

3.3.6 Interrupt mask set or clear register, SSPIMSC

The SSPIMSC register is the interrupt mask set or clear register. It is a read/write register.

On a read this register gives the current value of the mask on the relevant interrupt. A write of 1 to the particular bit sets the mask, enabling the interrupt to be read. A write of 0 clears the corresponding mask.

All the bits are cleared to 0 when reset. Table 3-7 shows the bit assignment of the SSPIMSC register.

Table 3-7 SSPIMSC register

Bits	Name	Type	Function
15:4	Reserved	-	Reserved, read as zero, do not modify.
3	TXIM	Read/write	Transmit FIFO interrupt mask: 0 = Tx FIFO half empty or less condition interrupt is masked 1 = Tx FIFO half empty or less condition interrupt is not masked.
2	RXIM	Read/write	Receive FIFO interrupt mask: 0 = Rx FIFO half full or less condition interrupt is masked 1 = Rx FIFO half full or less condition interrupt is not masked.
1	RTIM	Read/write	Receive timeout interrupt mask: 0 = Rx FIFO not empty and no read prior to timeout period interrupt is masked 1 = Rx FIFO not empty and no read prior to timeout period interrupt is not masked.
0	RORIM	Read/write	Receive overrun interrupt mask: 0 = Rx FIFO written to while full condition interrupt is masked 1 = Rx FIFO written to while full condition interrupt is not masked.

3.3.7 Raw interrupt status register, SSPRIS

The SSPRIS register is the raw interrupt status register. It is a read-only register. On a read this register gives the current raw status value of the corresponding interrupt prior to masking. A write has no effect. Table 3-8 shows the bit assignment of the SSPRIS register.

Table 3-8 SSPRIS register

Bits	Name	Type	Function
15:4	Reserved	-	Reserved, read as zero, do not modify
3	TXRIS	Read	Gives the raw interrupt state (prior to masking) of the SSPTXINTR interrupt
2	RXRIS	Read	Gives the raw interrupt state (prior to masking) of the SSPRXINTR interrupt
1	RTRIS	Read	Gives the raw interrupt state (prior to masking) of the SSPRTINTR interrupt
0	RORRIS	Read	Gives the raw interrupt state (prior to masking) of the SSPRORINTR interrupt

3.3.8 Masked interrupt status register, SSPMIS

The SSPMIS register is the masked interrupt status register. It is a read-only register. On a read this register gives the current masked status value of the corresponding interrupt. A write has no effect. Table 3-9 shows the bit assignment of the SSPMIS register.

Table 3-9 SSPMIS register

Bits	Name	Type	Function
15:4	Reserved	-	Reserved, read as zero, do not modify
3	TXMIS	Read	Gives the transmit FIFO masked interrupt state (after masking) of the SSPTXINTR interrupt
2	RXMIS	Read	Gives the receive FIFO masked interrupt state (after masking) of the SSPRXINTR interrupt
1	RTMIS	Read	Gives the receive timeout masked interrupt state (after masking) of the SSPRTINTR interrupt
0	RORMIS	Read	Gives the receive over run masked interrupt status (after masking) of the SSPRORINTR interrupt

3.3.9 Interrupt clear register, SSPICR

The SSPICR register is the interrupt clear register and is write-only. On a write of 1, the corresponding interrupt is cleared. A write of 0 has no effect. Table 3-10 shows the bit assignment of the SSPICR register.

Table 3-10 SSPICR register

Bits	Name	Type	Function
15:2	Reserved	-	Reserved, read as zero, do not modify
1	RTIC	Write	Clears the SSPRTINTR interrupt
0	RORIC	Write	Clears the SSPRORINTR interrupt

3.3.10 DMA control register, SSPDMACR

The SSPDMACR register is the DMA control register. It is a read/write register. All the bits are cleared to 0 on reset. Table 3-11 shows the bit assignment of the SSPDMACR register.

Table 3-11 SSPDMACR register

Bits	Name	Type	Function
15:2	Reserved	-	Reserved, read as zero, do not modify
1	Transmit DMA Enable (TXDMAE)	Read/write	If this bit is set to 1, DMA for the transmit FIFO is enabled
0	Receive DMA Enable (RXDMAE)	Read/write	If this bit is set to 1, DMA for the receive FIFO is enabled

3.3.11 Peripheral identification registers, SSPPeriphID0-3

The SSPPeriphID0-3 registers are four 8-bit registers, that span address locations 0xFE0 to 0xFEC. The registers can conceptually be treated as a single 32-bit register. The read-only registers provide the following options for the peripheral:

PartNumber[11:0]

This is used to identify the peripheral. The three digits product code 0x022 is used.

Designer ID[19:12]

This is the identification of the designer. ARM Ltd is 0x41 (ASCII A).

Revision[23:20]

This is the revision number of the peripheral. The number starts from 0 and is revision dependent.

Configuration[31:24]

This is the configuration option of the peripheral. The configuration value is 0.

Figure 3-1 shows the bit assignment for the SSPPeriphID0-3 registers.

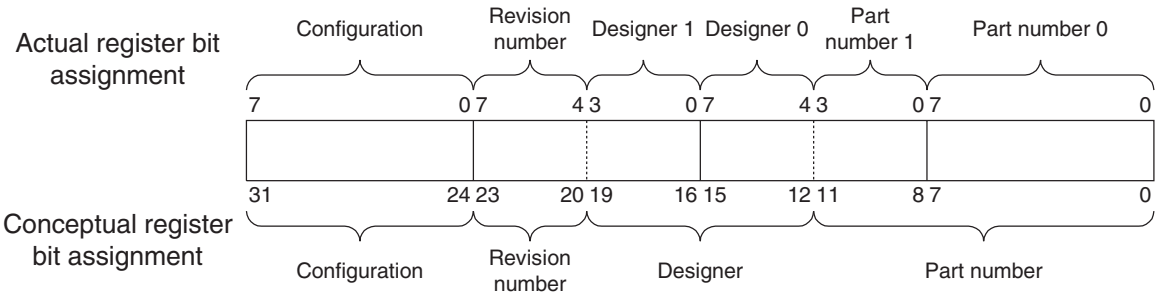


Figure 3-1 Peripheral identification register bit assignment

Note

When you design a systems memory map you must remember that the register has a 4KB-memory footprint. All memory accesses to the peripheral identification registers must be 32-bit, using the LDR and STR instructions.

The four 8-bit peripheral identification registers are described in the following subsections:

- *SSPPeriphID0 register*
- *SSPPeriphID1 register*
- *SSPPeriphID2 register* on page 3-16
- *SSPPeriphID3 register* on page 3-16.

SSPPeriphID0 register

The SSPPeriphID0 register is hard-coded and the fields within the register determine the reset value. Table 3-12 shows the bit assignment of the SSPPeriphID0 register.

Table 3-12 SSPPeriphID0 register

Bits	Name	Description
15:8	Reserved	Reserved, read undefined must read as zeros
7:0	PartNumber0	These bits read back as 0x22

SSPPeriphID1 register

The SSPPeriphID1 register is hard-coded and the fields within the register determine the reset value. Table 3-13 shows the bit assignment of the SSPPeriphID1 register.

Table 3-13 SSPPeriphID1 register

Bits	Name	Description
15:8	Reserved	Reserved, read undefined, must read as zeros
7:4	Designer0	These bits read back as 0x1
3:0	PartNumber1	These bits read back as 0x0.

SSPPeriphID2 register

The SSPPeriphID2 register is hard-coded and the fields within the register determine the reset value. Table 3-14 shows the bit assignment of the SSPPeriphID2 register.

Table 3-14 SSPPeriphID2 register

Bits	Name	Description
15:8	Reserved	Reserved, read undefined, must read as zeros
7:4	Revision	These bits return the peripheral revision
3:0	Designer1	These bits read back as 0x4

SSPPeriphID3 register

The SSPPeriphID3 register is hard-coded and the fields within the register determine the reset value. Table 3-15 shows the bit assignment of the SSPPeriphID3 register.

Table 3-15 SSPPeriphID3 register

Bits	Name	Description
15:8	Reserved	Reserved, read undefined, must read as zeros
7:0	Configuration	These bits read back as 0x00

3.3.12 PrimeCell identification registers, SSPPCellID0-3

The SSPPCellID0-3 registers are four 8-bit wide registers, that span address locations 0xFF0-0xFFC. The registers can conceptually be treated as a 32-bit register. The register is used as a standard cross-peripheral identification system. The SSPPCellID register is set to 0xB105F00D. Figure 3-2 on page 3-17 shows the bit assignment for the SSPPCellID0-3 registers.

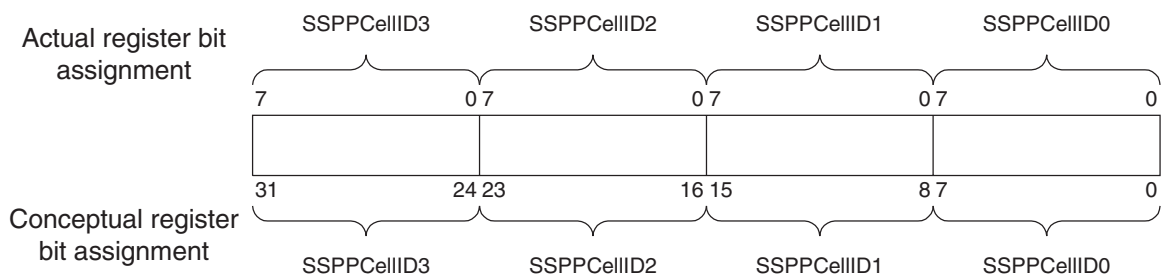


Figure 3-2 PrimeCell identification register bit assignment

The four, 8-bit PrimeCell identification registers are described in the following subsections:

- *SSPPCellIID0 register*
- *SSPPCellIID1 register* on page 3-18
- *SSPPCellIID2 register* on page 3-18
- *SSPPCellIID3 register* on page 3-18.

SSPPCellIID0 register

The SSPPCellIID0 register is hard-coded and the fields within the register determine the reset value. Table 3-16 shows the bit assignment of the SSPPCellIID0 register.

Table 3-16 SSPPCellIID0 register

Bits	Name	Description
15:8	Reserved	Reserved, read undefined, must read as zeros
7:0	SSPPCellIID0	These bits read back as 0x00

SSPPCellID1 register

The SSPPCellID1 register is hard-coded and the fields within the register determine the reset value. Table 3-17 shows the bit assignment of the SSPPCellID1 register.

Table 3-17 SSPPCellID1 register read bits

Bits	Name	Description
15:8	Reserved	Reserved, read undefined, must read as zeros
7:0	SSPPCellID1	These bits read back as 0xF0

SSPPCellID2 register

The SSPPCellID2 register is hard-coded and the fields within the register determine the reset value. Table 3-18 shows the bit assignment of the SSPPCellID2 register.

Table 3-18 SSPPCellID2 register read bits

Bits	Name	Description
15:8	Reserved	Reserved, read undefined, must read as zeros
7:0	SSPPCellID2	These bits read back as 0x05

SSPPCellID3 register

The SSPPCellID3 register is hard-coded and the fields within the register determine the reset value. Table 3-19 shows the bit assignment of the SSPPCellID3 register.

Table 3-19 SSPPCellID3 register read bits

Bits	Name	Description
15:8	Reserved	Reserved, read undefined, must read as zeros
7:0	SSPPCellID3	These bits read back as 0xB1

3.4 Interrupts

There are five interrupts generated by the PrimeCell SSP. Four of these are individual, maskable, active HIGH interrupts:

SSPRXINTR

PrimeCell SSP receive FIFO service interrupt request. See *SSPRXINTR*.

SSPTXINTR

PrimeCell SSP transmit FIFO service interrupt request. See *SSPTXINTR* on page 3-20.

SSPRORINTR

PrimeCell SSP receive overrun interrupt request. See *SSPRORINTR* on page 3-20.

SSPRTINTR

PrimeCell SSP time out interrupt request. See *SSPRTINTR* on page 3-20.

The fifth is a combined single interrupt **SSPINTR**. See *SSPINTR* on page 3-20.

You can mask each of the four individual maskable interrupts by setting the appropriate bits in the SSPIMSC register. Setting the appropriate mask bit HIGH enables the interrupt.

Provision of the individual outputs as well as a combined interrupt output, allows use of either a global interrupt service routine, or modular device drivers to handle interrupts.

The transmit and receive dynamic dataflow interrupts **SSPTXINTR** and **SSPRXINTR** have been separated from the status interrupts, so that data can be read or written in response to just the FIFO trigger levels.

The status of the individual interrupt sources can be read from SSPRIS and SSPMIS registers.

3.4.1 SSPRXINTR

The receive interrupt is asserted when there are four or more valid entries in the receive FIFO.

3.4.2 SSPTXINTR

The transmit interrupt is asserted when there are four or less valid entries in the transmit FIFO. The transmitter interrupt **SSPTXINTR** is not qualified with the PrimeCell SSP enable signal, which allows operation in one of two ways. Data can be written to the transmit FIFO prior to enabling the PrimeCell SSP and the interrupts. Alternatively, the PrimeCell SSP and interrupts can be enabled so that data can be written to the transmit FIFO by an interrupt service routine.

3.4.3 SSPRORINTR

The receive overrun interrupt **SSPORINTR** is asserted when the FIFO is already full and an additional data frame is received, causing an overrun of the FIFO. Data is over-written in the receive shift register, but not the FIFO.

3.4.4 SSPRTINTR

The receive timeout interrupt is asserted when the receive FIFO is not empty and the PrimeCell SSP has remained idle for a fixed 32 bit period. This mechanism ensures that the user is aware that data is still present in the receive FIFO and requires servicing. This interrupt is deasserted if the receive FIFO becomes empty by subsequent reads, or if new data is received on **SSPRXD**. It can also be cleared by writing to the **RTIC** bit in the **SSPICR** register.

3.4.5 SSPINTR

The interrupts are also combined into a single output **SSPINTR**, that is an OR function of the individual masked sources. You can connect this output to the system interrupt controller to provide another level of masking on an individual per-peripheral basis.

The combined PrimeCell SSP interrupt is asserted if any of the four individual interrupts above are asserted and enabled.

Chapter 4

Programmer's Model for Test

This chapter describes the additional logic for integration testing. It contains the following sections:

- *PrimeCell SSP test harness overview* on page 4-2
- *Scan testing* on page 4-3
- *Test registers* on page 4-4
- *Integration testing of block inputs* on page 4-8
- *Integration testing of block outputs* on page 4-10
- *Integration test summary* on page 4-13.

4.1 PrimeCell SSP test harness overview

The additional logic for integration vectors allows:

capture of input signals to the block

stimulation of the output signals.

The integration vectors provide a way of verifying that the PrimeCell SSP is correctly wired into a system. This is done by separately testing three groups of signals:

AMBA signals These are tested by checking the connections of all the address and data bits.

Primary input/output signals

These are tested using a simple trickbox that can demonstrate the correct connection of the input/output signals to external pads.

Intra-chip signals (such as interrupt sources)

The tests for these signals are system-specific, and enable you to write the necessary tests. Additional logic is implemented allowing you to read and write to each intra-chip input/output signal.

These test features are controlled by test registers. This allows you to test the PrimeCell SSP in isolation from the rest of the system using only transfers from the AMBA APB.

Off-chip test vectors are supplied using a 32-bit parallel *External Bus Interface* (EBI) and converted to internal AMBA bus transfers. The application of test vectors is controlled through the *Test Interface Controller* (TIC) AMBA bus master module.

4.2 Scan testing

The PrimeCell SSP has been designed to simplify:

- insertion of scan test cells
- use of *Automatic Test Pattern Generation* (ATPG).

This is the recommended method of manufacturing test.

4.3 Test registers

The PrimeCell SSP test registers are memory-mapped as shown in Table 4-1.

Table 4-1 Test registers memory map

Address	Type	Width	Reset value	Name	Description
SSP Base + 0x080	Read/write	2	0x0	SSPTCR	Test control register
SSP Base + 0x084	Read/write	5	0x00	SSPITIP	Integration test input register
SSP Base + 0x088	Read/write	14	0x0000	SSPITOP	Integration test output register
SSP Base + 0x08C	Read/write	16	0x0000	SSPTDR	Test data register

Each register shown in Table 4-1 is described in the following sections:

- *Test control register, SSPTCR*
- *Integration test input register, SSPITIP* on page 4-5
- *Integration test output register, SSPITOP* on page 4-5
- *Test data register, SSPTDR* on page 4-7.

4.3.1 Test control register, SSPTCR

SSPTCR is the test control register. This general test register controls operation of the PrimeCell SSP under test conditions. Table 4-2 shows the bit assignments for the SSPTCR.

Table 4-2 SSPTCR register bits

Bits	Name	Description
16:2	-	Reserved, unpredictable when read.
1	Test fifo enable (TESTFIFO)	When this bit is 1, a write to the SSPTDR writes data into the receive FIFO, and reads from the SSPTDR reads data out of the transmit FIFO. When this bit is 0, data cannot be read directly from the transmit FIFO or written directly to the receive FIFO (normal operation). The reset value is 0.
0	ITEN	Integration test enable. When this bit is 1, the PrimeCell SSP is placed in integration test mode, otherwise it is in normal mode.

4.3.2 Integration test input register, SSPITIP

SSPITIP is the integration test input register. It is a read/write register. In integration test mode it allows inputs to be both written to and read from. Table 4-3 shows the bit assignments for the SSPITIP.

Table 4-3 SSPITIP register bits

Bits	Name	Description
16:5	-	Reserved, unpredictable when read.
4	SSPTXDMACLR	Writes to this bit specify the value to be driven on the intra-chip input, SSPTXDMACLR , in the integration test mode. Reads return the value of SSPTXDMACLR at the output of the test multiplexor.
3	SSPRXDMACLR	Writes to this bit specify the value to be driven on the intra-chip input, SSPRXDMACLR , in the integration test mode. Reads return the value of SSPRXDMACLR at the output of the test multiplexor.
2	SSPCLKIN	Reads return the value of the SSPCLKIN primary input.
1	SSPFSSIN	Reads return the value of the SSPFSSIN primary input.
0	SSPRXD	Reads return the value of the SSPRXD primary input.

4.3.3 Integration test output register, SSPITOP

SSPITOP is the integration test output register. The primary outputs are hard-coded and the intra-chip outputs are read/write. In integration test mode it allows outputs to be both written to and read from. Table 4-4 shows the bit assignments for the SSPITOP.

Table 4-4 SSPITOP register bits

Bits	Name	Description
13	SSPTXDMASREQ	Intra-chip output. Writes specify the value to be driven on the SSPTXDMASREQ line in the integration test mode. Reads return the value of SSPTXDMASREQ at the output of the test multiplexor.
12	SSPTXDMABREQ	Intra-chip output. Writes specify the value to be driven on the SSPTXDMABREQ line in the integration test mode. Reads return the value of SSPTXDMABREQ at the output of the test multiplexor.
11	SSPRXDMASREQ	Intra-chip output. Writes specify the value to be driven on the SSPRXDMASREQ line in the integration test mode. Reads return the value of SSPRXDMASREQ at the output of the test multiplexor.

Table 4-4 SSPITOP register bits (continued)

Bits	Name	Description
10	SSPRXDMABREQ	Intra-chip output. Writes specify the value to be driven on the SSPRXDMABREQ line in the integration test mode. Reads return the value of SSPRXDMABREQ at the output of the test multiplexor.
9	SSPINTR	Intra-chip output. Writes specify the value to be driven on the SSPINTR line in the integration test mode. Reads return the value of SSPINTR at the output of the test multiplexor.
8	SSPTXINTR	Intra-chip output. Writes specify the value to be driven on the SSPTXINTR line in the integration test mode. Reads return the value of SSPTXINTR at the output of the test multiplexor.
7	SSPRXINTR	Intra-chip output. Writes specify the value to be driven on the SSPRXINTR line in the integration test mode. Reads return the value of SSPRXINTR at the output of the test multiplexor.
6	SSPRTINTR	Intra-chip output. Writes specify the value to be driven on the SSPRTINTR line in the integration test mode. Reads return the value of SSPRTINTR at the output of the test multiplexor.
5	SSPRORINTR	Intra-chip output. Writes specify the value to be driven on the SSPRORINTR line in the integration test mode. Reads return the value of SSPRORINTR at the output of the test multiplexor.
4	nSSPCTLOE	Primary output. Writes specify the value to be driven on the nSSPCTLOE line in the integration test mode.
3	nSSPOE	Primary output. Writes specify the value to be driven on the nSSPOE line in the integration test mode.
2	SSPCLKOUT	Primary output. Writes specify the value to be driven on the SSPCLKOUT line in the integration test mode.
1	SSPFSSOUT	Primary output. Writes specify the value to be driven on the SSPFSSOUT line in the integration test mode.
0	SSPTXD	Primary output. Writes specify the value to be driven on the SSPTXD line in the integration test mode.

4.3.4 Test data register, SSPTDR

SSPTDR is the test data register. It allows data to be written into the receive FIFO and read out from the transmit FIFO for test purposes. This test function is enabled by the **TESTFIFO** signal, bit 1 of the test control register (SSPTCR). Table 4-5 shows the bit assignments for the SSPTDR.

Table 4-5 SSPTDR register bits

Bits	Name	Description
31:16	-	Reserved, unpredictable when read.
15:0	DATA	When the TESTFIFO signal is asserted, data is written into the receive FIFO and read out of the transmit FIFO.

4.4 Integration testing of block inputs

The following sections describe the integration testing for the block inputs:

- *Intra-chip inputs*
- *Primary inputs* on page 4-9.

4.4.1 Intra-chip inputs

Figure 4-1 explains the implementation details of the input integration test harness. The ITEN bit is used as the control bit for the multiplexor, which is used in the read path of the **SSPTXDMACLR** and **SSPRXDMACLR** intra-chip inputs. If the ITEN control bit is deasserted, the **SSPTXDMACLR** and **SSPRXDMACLR** intra-chip inputs are routed as the internal **SSPTXDMACLR** and **SSPRXDMACLR** inputs respectively, otherwise the stored register values are driven on the internal line. All other hard-coded bits in the SSPITIP register are connected directly to the primary input pins.

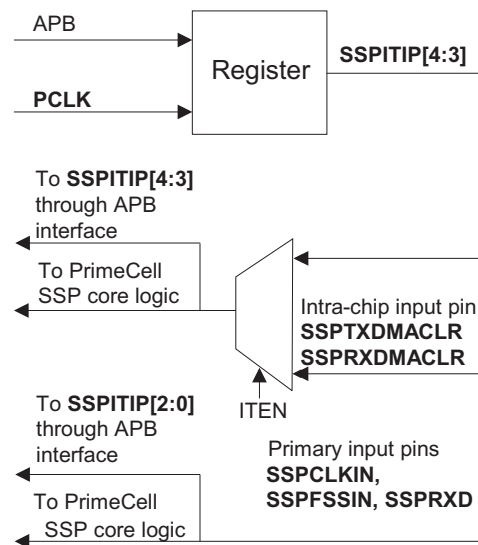


Figure 4-1 Input integration test harness

When you run integration tests with the PrimeCell SSP in a standalone test setup:

- Write a 1 to the ITEN bit in the control register. This selects the test path from the SSPITIP[1:0] register bits to the **SSPRXDMACLR** and **SSPTXDMACLR** signals.

- Write a 1 and then a 0 to each of the SSPITIP[4:3] register bits, and read the same register bits to ensure that the value written is read out.

When you run integration tests with the PrimeCell SSP as part of an integrated system:

- Write a 0 to the ITEN bit in the control register. This selects the normal path from the external **SSPRXDMACLR** pin to the internal **SSPRXDMACLR** signal, and the path from the external **SSPTXDMACLR** pin to the internal **SSPTXDMACLR** pin.
- Write a 1 and then a 0 to the internal test registers of the DMA controller to toggle the **SSPRXDMACLR** signal connection between the DMA controller and the PrimeCell SSP. Read from the SSPITIP[3] register bit to verify that the value written into the DMA controller, is read out through the PrimeCell SSP. Similarly, write a 1 and then a 0 to the internal registers of the DMA controller to toggle the **SSPTXDMACLR** signal connection between the DMA controller and the PrimeCell SSP. Read from the SSPITIP[4] register bit to verify that the value written into the DMA controller, is read out through the PrimeCell SSP.

4.4.2 Primary inputs

The following primary inputs are tested using the integration vector trickbox, by looping back the primary inputs as follows:

- **SSPTXD** to **SSPRXD**
- **SSPCLKOUT** to **SSPCLKIN**
- **SSPFSSOUT** to **SSPFSSIN**.

Write a 1 to the ITEN bit in the SSPTCR control register. 1s and 0s are driven onto the primary output lines through the SSPITOP[4:0] register bits, which include the enable signals, and read back through the SSPITIP[2:0] register bits.

4.5 Integration testing of block outputs

The following sections describe the integration testing for the block outputs:

- *Intra-chip outputs*
- *Primary outputs* on page 4-11.

4.5.1 Intra-chip outputs

Use this test for the following outputs:

- **SSPTXDMASREQ**
- **SSPTXDMABREQ**
- **SSPRXDMASREQ**
- **SSPRXDMABREQ**
- **SSPINTR**
- **SSPTXINTR**
- **SSPRXINTR**
- **SSPTINTR**
- **SSPRORINTR**.

When you run integration tests with the PrimeCell SSP in a standalone test setup:

- Write a 1 to the ITEN bit in the control register. This selects the test path from the SSPITOP[13:5] register bits to the intra-chip output signals.
- Write a 1 and then a 0 to the SSPITOP[13:5] register bits, and read the same register bits to verify that the value written is read out.

When you run integration tests with the PrimeCell SSP as part of an integrated system:

- Write a 1 to the ITEN bit in the control register. This selects the test path from the SSPITOP[13:5] register bits to the intra-chip output signals.
- Write a 1 and then a 0 to the SSPITOP[13:5] register bits to toggle the signal connections between the DMA controller/interrupt controller and the PrimeCell SSP. Read from the internal test registers of the DMA controller/interrupt controller to verify that the value written into the SSPITOP[13:5] register bits is read out through the PrimeCell SSP.

Figure 4-4 on page 4-12 explains the implementation details of the output integration test harness for intra-chip outputs.

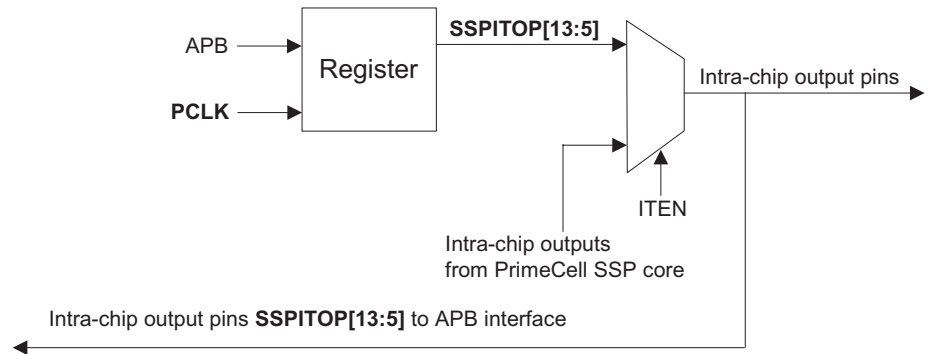


Figure 4-2 Output integration test harness, intra-chip outputs

4.5.2 Primary outputs

Integration testing of primary outputs and primary inputs is carried out using the integration vector trickbox. Use this test for the following outputs:

- **SSPTXD**
- **SSPCLKOUT**
- **SSPFSSOUT**
- **nSSPCTLOE**
- **nSSPOE**.

Note

Only the **SSPTXD**, **SSPCLKOUT**, and **SSPFSSOUT** signals are available at the output pads of a typical configuration. The **nSSPOE** and **nSSPCTLOE** signals are internal connections to the pad.

Verify the primary input and output pin connections as follows:

The primary outputs, **SSPTXD**, **SSPCLKOUT**, and **SSPFSSOUT** are directly connected to **SSPRXD**, **SSPCLKIN**, and **SSPFSSIN** respectively by the integration trickbox shown in Figure 4-3 on page 4-12. To test the **nSSPOE** and **nSSPCTLOE** connections, you can apply a weak pull down/pull to the tristate pins through the trickbox.

- All the primary outputs can be accessed through the SSPITOP register. Different data patterns are written to the output pins using the SSPITOP register.

- The looped-back data is read back through the SSPITIP register.

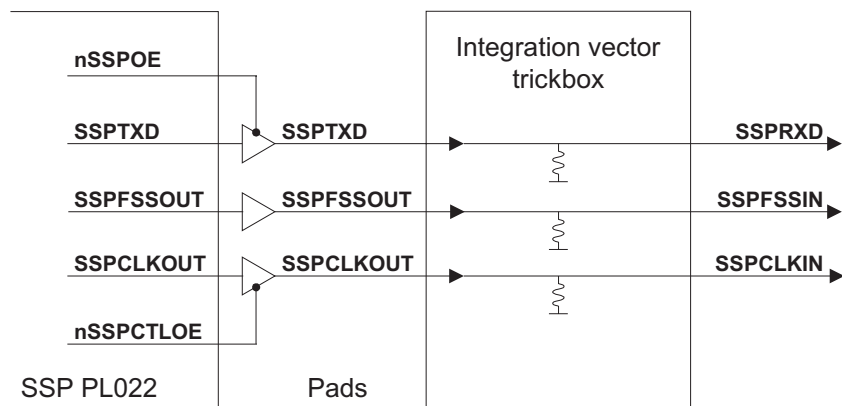


Figure 4-3 Primary outputs routed to primary inputs

Figure 4-4 shows implementation details of the output integration test harness in the case of primary outputs.

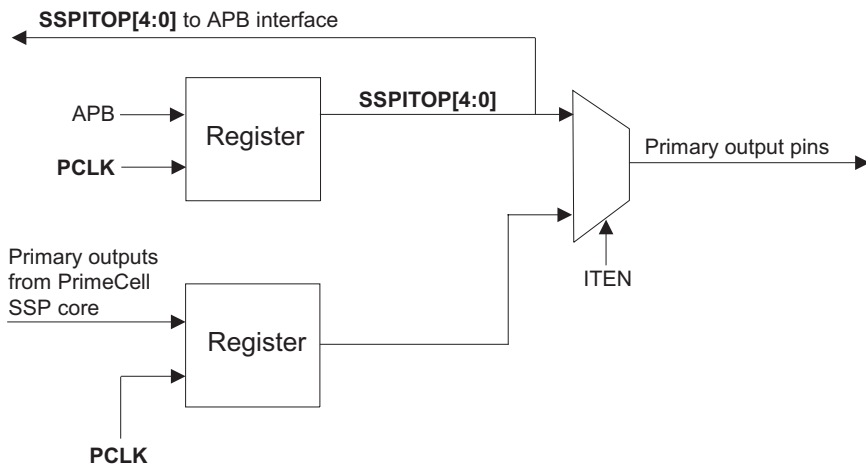


Figure 4-4 Output integration test harness, primary outputs

4.6 Integration test summary

Table 4-6 summarizes the integration test strategy for all PrimeCell SSP pins.

Table 4-6 PrimeCell SSP integration test strategy

Name	Type	Source/ destination	Test strategy
PRESETn	Input	Reset controller	Not tested using integration test vectors
PADDR [11:2]	Input	APB	Register read/write
PCLK	Input	APB	Register read/write
PENABLE	Input	APB	Register read/write
PRDATA [15:0]	Output	APB	Register read/write
PSEL	Input	APB	Register read/write
PWDATA [15:0]	Input	APB	Register read/write
PWRITE	Input	APB	Register read/write
SSPCLK	Input	Clock generator	Not tested using integration test vectors
nSSPRST	Input	Reset controller	Not tested using integration test vectors
SSPINTR	Output	Interrupt controller	Using SSPITOP register
SSPRXINTR	Output	Interrupt controller	Using SSPITOP register
SSPTXINTR	Output	Interrupt controller	Using SSPITOP register
SSPRTINTR	Output	Interrupt controller	Using SSPITOP register
SSPRORINTR	Output	Interrupt controller	Using SSPITOP register
SSPTXDMASREQ	Output	DMA controller	Using SSPITOP register
SSPRXDMASREQ	Output	DMA controller	Using SSPITOP register
SSPTXDMABREQ	Output	DMA controller	Using SSPITOP register
SSPRXDMABREQ	Output	DMA controller	Using SSPITOP register
SSPTXDMACLR	Input	DMA controller	Using SSPITIP register
SSPRXDMACLR	Input	DMA controller	Using SSPITIP register
SCANENABLE	Input	Test controller	Not tested using integration test vectors
SCANINPCLK	Input	Test controller	Not tested using integration test vectors

Table 4-6 PrimeCell SSP integration test strategy (continued)

Name	Type	Source/ destination	Test strategy
SCANINSSPCLK	Input	Test controller	Not tested using integration test vectors
SCANOUTPCLK	Output	Test controller	Not tested using integration test vectors
SCANOUTSSPCLK	Output	Test controller	Not tested using integration test vectors
SSPRXD	Input	PAD	Using integration vector trickbox and SSPITIP and SSPITOP registers
SSPFSSIN	Input	PAD	Using integration vector trickbox and SSPITIP and SSPITOP registers
SSPCLKIN	Input	PAD	Using integration vector trickbox and SSPITIP and SSPITOP registers
SSPTXD	Output	PAD	Using integration vector trickbox and SSPITIP and SSPITOP registers
SSPFSSOUT	Output	PAD	Using integration vector trickbox and SSPITIP and SSPITOP registers
SSPCLKOUT	Output	PAD	Using integration vector trickbox and SSPITIP and SSPITOP registers
nSSPCTLOE	Output	PAD	Using integration vector trickbox and SSPITIP and SSPITOP registers
nSSPOE	Output	PAD	Using integration vector trickbox and SSPITIP and SSPITOP registers

Appendix A

Signal Descriptions

This appendix describes the signals that interface with the ARM PrimeCell SSP (PL022). It contains the following sections:

- *AMBA APB signals* on page A-2
- *On-chip signals* on page A-3
- *Signals to pads* on page A-5.

A.1 AMBA APB signals

The PrimeCell SSP module is connected to the AMBA APB as a bus slave. Table A-1 describes the APB signals that are used and produced.

Table A-1 AMBA APB signal descriptions

Name	Type	Source/destination	Description
PRESETn	Input	Reset controller	Bus reset signal, active LOW.
PADDR[11:2]	Input	APB bridge	Subset of AMBA APB address bus.
PCLK	Input	Clock generator	AMBA APB clock, used to time all bus transfers.
PENABLE	Input	APB bridge	AMBA APB enable signal. PENABLE is asserted HIGH for one cycle of PCLK to enable a bus transfer.
PRDATA[15:0]	Output	APB bridge	Unidirectional AMBA APB read data bus.
PSEL	Input	APB bridge	PrimeCell SSP select signal from decoder. When set to 1 this signal indicates the slave device is selected by the AMBA APB bridge, and that a data transfer is required.
PWDATA[15:0]	Input	APB bridge	Unidirectional AMBA APB write data bus.
PWRITE	Input	APB bridge	AMBA APB transfer direction signal, indicates a write access when HIGH, read access when LOW.

A.2 On-chip signals

The reset inputs are asynchronously asserted but synchronously removed for each of the clock domains within the PrimeCell SSP. This ensures that logic is reset even if clocks are not present, to avoid any static power consumption problems at power up. Each clock domain has a individual reset to simplify the process of inserting scan test cells.

Table A-2 shows the non-AMBA signals from the block.

Table A-2 On-chip signals

Name	Type	Source/ destination	Description
SSPCLK	Input	Clock generator	Main PrimeCell SSP clock input.
nSSPRST	Input	Reset controller	PrimeCell SSP reset signal to SSPCLK clock domain, active LOW. The reset controller must use PRESETn to assert nSSPRST asynchronously, but negate it synchronously with SSPCLK .
SSPTXINTR	Output	Interrupt controller	Transmit FIFO service request.
SSPRXINTR	Output	Interrupt controller	Receive FIFO service request.
SSPRORINTR	Output	Interrupt controller	PrimeCell SSP receive overrun interrupt.
SSPRTINTR	Output	Interrupt controller	PrimeCell SSP receive timeout interrupt.
SSPINTR	Output	Interrupt controller	PrimeCell SSP interrupt. This interrupt is an OR of the four individual interrupts SSPTXINTR , SSPRXINTR , SSPRTINTR , and SSPRORINTR .
SSPTXDMASREQ	Output	DMA controller	PrimeCell SSP transmit DMA single request (active HIGH).
SSPRXDMASREQ	Output	DMA controller	PrimeCell SSP receive DMA single request (active HIGH).
SSPTXDMABREQ	Output	DMA controller	PrimeCell SSP transmit DMA burst request (active HIGH).
SSPRXDMABREQ	Output	DMA controller	PrimeCell SSP receive DMA burst request (active HIGH).
SSPTXDMACLR	Input	DMA controller	DMA request clear, asserted by the DMA controller to clear the transmit request signals. If DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.
SSPRXDMACLR	Input	DMA controller	DMA request clear, asserted by the DMA controller to clear the receive request signals. If DMA burst transfer is requested, the clear signal is asserted during the transfer of the last data in the burst.

Table A-2 On-chip signals (continued)

Name	Type	Source/ destination	Description
SCANENABLE	Input	Test controller	Place holder for scan path select signal.
SCANINPCLK	Input	Test controller	Place holder for scan data input signal (PCLK domain).
SCANOUTPCLK	Output	Test controller	Place holder for scan data output signal (PCLK domain).
SCANINSSPCLK	Input	Test controller	Place holder for scan data input signal (SSPCLK domain).
SCANOUTSSPCLK	Output	Test controller	Place holder for scan data output signal (SSPCLK domain).

A.3 Signals to pads

Table A-3 describes the signals from the PrimeCell SSP to input/output pads of the chip. It is your responsibility to make proper use of the peripheral pins to meet the exact interface requirements.

Table A-3 Pad signal descriptions

Name	Type	Source/ destination	Description
SSPFSSOUT	Output	Pad	PrimeCell SSP frame, or slave select output (master).
SSPCLKOUT	Output	Pad	PrimeCell SSP clock output (master).
SSPRXD	Input	Pad	PrimeCell SSP receive data input.
SSPTXD	Output	Pad	PrimeCell SSP transmit data output.
nSSPCTLOE	Output	Pad	Output enable signal (active LOW) for SSPCLKOUT output from the PrimeCell SSP. This output is cleared when the device is in master mode and set when the device is in slave mode.
SSPFSSIN	Input	Pad	PrimeCell SSP frame input (slave).
SSPCLKIN	Input	Pad	PrimeCell SSP clock input (slave).
nSSPOE	Output	Pad	Output enable signal (active LOW) to indicate when SSPTXD is valid.

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